Chapter 2 Exercises

Solutions to Rick Durrett's
Probability: Theory and Examples

June 12, 2026

These solutions use the local Chapter 2 LLM/Viki knowledge set in `Probability/LLM_Viki_Dataset/Chapter2_Knowledge`. The cited knowledge IDs refer to entries in `chapter2_knowledge_pieces.jsonl`.

Exercise 2.1.1

Question. Suppose $(X_1, \dots, X_n)$ has density $f(x_1, \dots, x_n)$, that is

$$\mathbb{P}((X_1, \dots, X_n) \in A) = \int_A f(x)\, dx, \qquad A \in \mathcal{B}(\mathbb{R}^n).$$

If $f(x)$ can be written as $g_1(x_1) \cdots g_n(x_n)$, where the $g_m \geq 0$ are measurable, then $X_1, \dots, X_n$ are independent. Note that the g_m are not assumed to be probability densities.

Knowledge used. `durrett_ch2_joint_law_product_measure`; `durrett_ch2_pi_lambda_independence`; `durrett_ch2_rv_sigma_independence`.

Solution. Let

$$c_i = \int_{\mathbb{R}} g_i(x)\, dx.$$

By Tonelli's theorem,

$$1 = \int_{\mathbb{R}^n} f(x)\, dx = \prod_{i=1}^n \int_{\mathbb{R}} g_i(x_i)\, dx_i = \prod_{i=1}^n c_i.$$

Thus each c_i is finite and positive. Define $h_i = g_i / c_i$. Then each h_i is a probability density and

$$f(x_1, \dots, x_n) = \prod_{i=1}^n h_i(x_i).$$

For rectangles $A_1 \times \cdots \times A_n$,

$$\begin{aligned}\mathbb{P}(X_1 \in A_1, \dots, X_n \in A_n) &= \int_{A_1 \times \cdots \times A_n} \prod_{i=1}^n h_i(x_i)\, dx \\ &= \prod_{i=1}^n \int_{A_i} h_i(x_i)\, dx_i = \prod_{i=1}^n \mathbb{P}(X_i \in A_i).\end{aligned}$$

The rectangle class is a pi-system generating $\mathcal{B}(\mathbb{R}^n)$, so the factorization extends to the generated sigma-fields. Hence $X_1, \dots, X_n$ are independent.

Exercise 2.1.2

Question. Suppose $X_1, \dots, X_n$ are random variables that take values in countable sets $S_1, \dots, S_n$. Then in order for $X_1, \dots, X_n$ to be independent, it is sufficient that whenever $x_i \in S_i$,

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n) = \prod_{i=1}^{n} \mathbb{P}(X_i = x_i).$$

Knowledge used. `durrett_ch2_event_independence`; `durrett_ch2_pi_lambda_independence`; `durrett_ch2_joint_law_product_measure`.

Solution. Let $A_i \subseteq S_i$. Since the S_i are countable, Tonelli's theorem for nonnegative sums gives

$$\begin{aligned}\mathbb{P}(X_1 \in A_1, \dots, X_n \in A_n) &= \sum_{x_1 \in A_1} \cdots \sum_{x_n \in A_n} \mathbb{P}(X_1 = x_1, \dots, X_n = x_n) \\ &= \sum_{x_1 \in A_1} \cdots \sum_{x_n \in A_n} \prod_{i=1}^{n} \mathbb{P}(X_i = x_i) \\ &= \prod_{i=1}^{n} \sum_{x_i \in A_i} \mathbb{P}(X_i = x_i) = \prod_{i=1}^{n} \mathbb{P}(X_i \in A_i).\end{aligned}$$

Thus the defining product formula holds for arbitrary subsets of the countable state spaces, and $X_1, \dots, X_n$ are independent.

Exercise 2.1.3

Question. Let $\rho(x, y)$ be a metric.

(i) Suppose h is differentiable with $h(0) = 0$, $h'(x) > 0$ for $x > 0$, and $h'(x)$ decreasing on $[0, \infty)$. Then $h(\rho(x, y))$ is a metric.

(ii) $h(x) = x/(x + 1)$ satisfies the hypotheses in (i).

Knowledge used. `durrett_ch2_construct_independent_variables`.

Solution. Since $h'(x) > 0$ for $x > 0$ and $h(0) = 0$, $h(t) \ge 0$ for $t \ge 0$, and $h(t) = 0$ if and only if $t = 0$. Therefore $d(x, y) = h(\rho(x, y))$ is nonnegative and equals 0 exactly when $x = y$. Symmetry follows immediately from symmetry of ρ.

It remains to prove the triangle inequality. Since h' is decreasing, h is concave on $[0, \infty)$. For $a, b \ge 0$,

$$h(a + b) - h(a) = \int_a^{a+b} h'(u)\, du \le \int_0^b h'(u)\, du = h(b).$$

Thus $h(a + b) \le h(a) + h(b)$. Since $\rho(x, z) \le \rho(x, y) + \rho(y, z)$ and h is increasing,

$$h(\rho(x, z)) \le h(\rho(x, y) + \rho(y, z)) \le h(\rho(x, y)) + h(\rho(y, z)).$$

So $h \circ \rho$ is a metric.

For $h(x) = x/(x + 1)$,

$$h(0) = 0, \qquad h'(x) = \frac{1}{(x + 1)^2} > 0, \qquad h''(x) = -\frac{2}{(x + 1)^3} < 0.$$

Hence h' is decreasing and the hypotheses in (i) hold.

Exercise 2.1.4

Question. Let $\Omega = (0,1)$, $\mathcal{F}$ be the Borel sets, and $\mathbb{P}$ be Lebesgue measure. Let

$$X_n(\omega) = \sin(2\pi n\omega), \qquad n = 1, 2, \ldots.$$

Show that X_n are uncorrelated but not independent.

Knowledge used. durrett_ch2_expectation_factorization; durrett_ch2_rv_sigma_independence; durrett_ch2_event_independence.

Solution. For every n,

$$\mathbb{E}X_n = \int_0^1 \sin(2\pi n\omega)\, d\omega = 0.$$

If $m \neq n$, the trigonometric orthogonality identity gives

$$\mathbb{E}(X_m X_n) = \int_0^1 \sin(2\pi m\omega)\sin(2\pi n\omega)\, d\omega = 0.$$

Thus $\mathbb{E}(X_m X_n) = \mathbb{E}X_m\, \mathbb{E}X_n$ for $m \neq n$, so the variables are uncorrelated.

They are not independent. Indeed,

$$X_2^2 = \sin^2(4\pi\omega) = 4\sin^2(2\pi\omega)\big(1 - \sin^2(2\pi\omega)\big) = 4X_1^2(1 - X_1^2).$$

Thus X_2^2 is $\sigma(X_1)$-measurable. If X_1 and X_2 were independent, then X_2^2 would be independent of $\sigma(X_1)$ and hence independent of itself. A random variable independent of itself must be almost surely constant, but $4X_1^2(1 - X_1^2)$ is not almost surely constant. Therefore X_1 and X_2 are not independent, so the sequence is not independent.

Exercise 2.1.5

Question.

(i) Show that if X and Y are independent with distributions μ and ν, then

$$\mathbb{P}(X + Y = 0) = \sum_y \mu(\{-y\})\nu(\{y\}).$$

(ii) Conclude that if X has continuous distribution, then $\mathbb{P}(X = Y) = 0$ for independent X and Y.

Knowledge used. durrett_ch2_joint_law_product_measure; durrett_ch2_expectation_factorization; durrett_ch2_convolution_cdf.

Solution. By independence, the joint law of (X, Y) is $\mu \times \nu$. The set

$$\{(x, y) : x + y = 0\}$$

can have positive product measure only at atoms. More explicitly, for fixed y the vertical section contains the single point $x = -y$, so Tonelli gives

$$\mathbb{P}(X + Y = 0) = \int \mu(\{-y\})\, \nu(dy).$$

The integrand is nonzero only when $-y$ is an atom of μ. A probability measure has at most countably many atoms, so this integral is the countable sum

$$\mathbb{P}(X+Y=0)=\sum_y \mu(\{-y\})\nu(\{y\}).$$

For (ii), apply (i) to X and $-Y$. If X has continuous distribution, then $\mu(\{y\})=0$ for every y, so

$$\mathbb{P}(X=Y)=\mathbb{P}(X-Y=0)=0.$$

Exercise 2.1.6

Question. Prove directly from the definition that if X and Y are independent and f and g are measurable functions, then $f(X)$ and $g(Y)$ are independent.

Knowledge used. `durrett_ch2_functions_of_independent_blocks`; `durrett_ch2_rv_sigma_independence`.

Solution. Let $A,B\in\mathcal{B}(\mathbb{R})$. Since f and g are measurable, $f^{-1}(A)$ and $g^{-1}(B)$ are Borel sets. Therefore

$$\begin{aligned}\mathbb{P}(f(X)\in A,\ g(Y)\in B)&=\mathbb{P}(X\in f^{-1}(A),\ Y\in g^{-1}(B))\\&=\mathbb{P}(X\in f^{-1}(A))\mathbb{P}(Y\in g^{-1}(B))\\&=\mathbb{P}(f(X)\in A)\mathbb{P}(g(Y)\in B).\end{aligned}$$

This is exactly the definition of independence of $f(X)$ and $g(Y)$.

Exercise 2.1.7

Question. Let $K\geq 3$ be a prime and let X and Y be independent random variables uniformly distributed on $\{0,1,\dots,K-1\}$. For $0\leq n<K$, let

$$Z_n=X+nY \pmod K.$$

Show that $Z_0,Z_1,\dots,Z_{K-1}$ are pairwise independent. They are not independent because if we know the values of two of the variables then we know the values of all the variables.

Knowledge used. `durrett_ch2_event_independence`; `durrett_ch2_functions_of_independent_blocks`; `durrett_ch2_joint_law_product_measure`.

Solution. Fix $r\neq s$ in $\{0,\dots,K-1\}$. For any $a,b\in\{0,\dots,K-1\}$, the equations

$$X+rY\equiv a,\qquad X+sY\equiv b \pmod K$$

have a unique solution modulo K. Indeed, subtracting gives

$$(s-r)Y\equiv b-a \pmod K.$$

Since K is prime and $s-r\not\equiv 0 \pmod K$, $s-r$ is invertible modulo K, so Y is uniquely determined, and then X is uniquely determined. Since (X,Y) is uniform on K^2 pairs,

$$\mathbb{P}(Z_r=a,Z_s=b)=K^{-2}.$$

Also each Z_r is uniform on $\{0,\dots,K-1\}$, so

$$\mathbb{P}(Z_r=a)\mathbb{P}(Z_s=b)=K^{-1}K^{-1}=K^{-2}.$$

Thus Z_r and Z_s are independent. Since $r \neq s$ were arbitrary, the family is pairwise independent.

The whole family is not independent. Any two distinct Z_r, Z_s determine X and Y, hence determine all Z_n. For example,

$$\mathbb{P}(Z_0 = 0, \ldots, Z_{K-1} = 0) = \mathbb{P}(X = 0, Y = 0) = K^{-2},$$

whereas mutual independence would give K^{-K}. Since $K \geq 3$, these numbers are different.

Exercise 2.1.8

Question. Find four random variables taking values in $\{-1, 1\}$ so that any three are independent but all four are not. Hint: Consider products of independent random variables.

Knowledge used. durrett_ch2_event_independence; durrett_ch2_functions_of_independent_blocks; durrett_ch2_expectation_factorization.

Solution. Let U, V, W be independent random variables with

$$\mathbb{P}(U = 1) = \mathbb{P}(U = -1) = \frac{1}{2},$$

and similarly for V and W. Define

$$X_1 = U, \qquad X_2 = V, \qquad X_3 = W, \qquad X_4 = UVW.$$

Every X_i takes values in $\{-1, 1\}$.

The variables X_1, X_2, X_3 are independent by construction. Consider, for instance, X_1, X_2, X_4. If $a, b, c \in \{-1, 1\}$, then

$$\{X_1 = a, X_2 = b, X_4 = c\} = \{U = a, V = b, W = abc\},$$

which has probability $1/8$. Since each of X_1, X_2, X_4 is symmetric on $\{-1, 1\}$, this equals the product of the three marginal probabilities. The same argument applies to every triple.

However,

$$X_1 X_2 X_3 X_4 = U \cdot V \cdot W \cdot UVW = 1$$

almost surely. If all four variables were independent, then each of the 16 sign vectors would have probability $1/16$, including sign vectors whose product is -1. This is impossible. Hence any three are independent, but all four are not.

Exercise 2.1.9

Question. Let $\Omega = \{1, 2, 3, 4\}$, $\mathcal{F}$ be all subsets of Ω, and $\mathbb{P}(\{i\}) = 1/4$. Give an example of two collections of sets $\mathcal{A}_1$ and $\mathcal{A}_2$ that are independent but whose generated sigma-fields are not.

Knowledge used. durrett_ch2_pi_lambda_independence; durrett_ch2_event_independence.

Solution. Let

$$A = \{1, 2\}, \qquad B = \{1, 3\}, \qquad C = \{1, 4\},$$

and set

$$\mathcal{A}_1 = \{A, B\}, \qquad \mathcal{A}_2 = \{C\}.$$

Then

$$\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C) = \frac{1}{2},$$

and

$$\mathbb{P}(A \cap C) = \mathbb{P}(\{1\}) = \frac{1}{4} = \mathbb{P}(A)\mathbb{P}(C),$$

while

$$\mathbb{P}(B \cap C) = \mathbb{P}(\{1\}) = \frac{1}{4} = \mathbb{P}(B)\mathbb{P}(C).$$

Thus the two collections are independent.

However, $\sigma(\mathcal{A}_1)$ contains $\{1\} = A \cap B$, while $C \in \sigma(\mathcal{A}_2)$. These two events are not independent:

$$\mathbb{P}(\{1\} \cap C) = \frac{1}{4} \neq \frac{1}{8} = \mathbb{P}(\{1\})\mathbb{P}(C).$$

Therefore $\sigma(\mathcal{A}_1)$ and $\sigma(\mathcal{A}_2)$ are not independent. This illustrates why the pi-system hypothesis in the independence extension theorem is essential.

Exercise 2.1.10

Question. Show that if X and Y are independent, integer-valued random variables, then

$$\mathbb{P}(X + Y = n) = \sum_m \mathbb{P}(X = m)\mathbb{P}(Y = n - m).$$

Knowledge used. `durrett_ch2_joint_law_product_measure`; `durrett_ch2_convolution_cdf`; `durrett_ch2_expectation_factorization`.

Solution. The events $\{X = m, Y = n - m\}$, $m \in \mathbb{Z}$, are disjoint and their union is $\{X + Y = n\}$. Hence

$$\begin{aligned}\mathbb{P}(X + Y = n) &= \sum_{m \in \mathbb{Z}} \mathbb{P}(X = m, Y = n - m) \\ &= \sum_{m \in \mathbb{Z}} \mathbb{P}(X = m)\mathbb{P}(Y = n - m),\end{aligned}$$

where the last step uses independence.

Exercise 2.1.11

Question. In Example 1.6.13, the Poisson distribution with parameter λ is given by

$$\mathbb{P}(Z = k) = e^{-\lambda}\frac{\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots.$$

Use the previous exercise to show that if $X = \text{Poisson}(\lambda)$ and $Y = \text{Poisson}(\mu)$ are independent, then $X + Y = \text{Poisson}(\lambda + \mu)$.

Knowledge used. `durrett_ch2_convolution_cdf`; `durrett_ch2_joint_law_product_measure`; `durrett_ch2_standard_convolution_examples`.

Solution. For $n = 0, 1, 2, \ldots$, Exercise 2.1.10 gives

$$\begin{aligned}\mathbb{P}(X + Y = n) &= \sum_{m=0}^{n} e^{-\lambda}\frac{\lambda^m}{m!} e^{-\mu}\frac{\mu^{n-m}}{(n-m)!} \\ &= e^{-(\lambda+\mu)}\frac{1}{n!}\sum_{m=0}^{n}\binom{n}{m}\lambda^m\mu^{n-m} \\ &= e^{-(\lambda+\mu)}\frac{(\lambda+\mu)^n}{n!}.\end{aligned}$$

This is the probability mass function of a $\text{Poisson}(\lambda + \mu)$ random variable.

Exercise 2.1.12

Question. X is said to have a Binomial(n,p) distribution if

$$\mathbb{P}(X=m)=\binom{n}{m}p^m(1-p)^{n-m}.$$

(i) Show that if $X=$ Binomial(n,p) and $Y=$ Binomial(m,p) are independent, then $X+Y=$ Binomial$(n+m,p)$.

(ii) Look at Example 1.6.12 and use induction to conclude that the sum of n independent Bernoulli(p) random variables is Binomial(n,p).

Knowledge used. durrett_ch2_convolution_cdf; durrett_ch2_joint_law_product_measure; durrett_ch2_standard_convolution_examples.

Solution. For $k=0,\dots,n+m$, Exercise 2.1.10 gives

$$\begin{aligned}\mathbb{P}(X+Y=k)&=\sum_j\binom{n}{j}p^j(1-p)^{n-j}\binom{m}{k-j}p^{k-j}(1-p)^{m-k+j}\\&=p^k(1-p)^{n+m-k}\sum_j\binom{n}{j}\binom{m}{k-j}.\end{aligned}$$

By Vandermonde's identity,

$$\sum_j\binom{n}{j}\binom{m}{k-j}=\binom{n+m}{k},$$

so

$$\mathbb{P}(X+Y=k)=\binom{n+m}{k}p^k(1-p)^{n+m-k}.$$

Therefore $X+Y$ is Binomial$(n+m,p)$.

For (ii), a Bernoulli(p) variable is Binomial$(1,p)$. The case $n=1$ is immediate. If the sum of n independent Bernoulli(p) variables is Binomial(n,p), then adding one more independent Bernoulli(p) variable and applying part (i) gives a Binomial$(n+1,p)$ distribution. The conclusion follows by induction.

Exercise 2.1.13

Question. It should not be surprising that the distribution of $X+Y$ can be $F*G$ without the random variables being independent. Suppose $X,Y\in\{0,1,2\}$ and each takes each value with probability 1/3.

(a) Find the distribution of $X+Y$ assuming X and Y are independent.

(b) Find all the joint distributions (X,Y) so that the distribution of $X+Y$ is the same as the answer to (a).

Knowledge used. durrett_ch2_convolution_cdf; durrett_ch2_joint_law_product_measure; durrett_ch2_event_independence.

Solution. If X and Y are independent and uniform on $\{0, 1, 2\}$, then the number of pairs (x, y) with $x + y = k$ is $1, 2, 3, 2, 1$ for $k = 0, 1, 2, 3, 4$. Hence

$$\mathbb{P}(X + Y = k) = \begin{cases} 1/9, & k = 0, 4, \\ 2/9, & k = 1, 3, \\ 3/9, & k = 2. \end{cases}$$

For (b), write $p_{ij} = \mathbb{P}(X = i, Y = j)$ and scale by $q_{ij} = 9p_{ij}$. The marginal conditions say each row and each column of $q = (q_{ij})_{0 \le i,j \le 2}$ sums to 3. The required distribution of $X + Y$ says the anti-diagonal sums are $1, 2, 3, 2, 1$.

Solving these linear constraints gives the one-parameter family

$$q = \begin{pmatrix} 1 & t & 2-t \\ 2-t & 1 & t \\ t & 2-t & 1 \end{pmatrix}, \qquad 0 \le t \le 2.$$

Therefore all such joint distributions are

$$\mathbb{P}(X = i, Y = j) = \frac{1}{9} \begin{pmatrix} 1 & t & 2-t \\ 2-t & 1 & t \\ t & 2-t & 1 \end{pmatrix}_{ij}, \qquad 0 \le t \le 2.$$

The independent case corresponds to $t = 1$, but every $t \in [0, 2]$ gives the same distribution for $X + Y$.

Exercise 2.1.14

Question. Let $X, Y \ge 0$ be independent with distribution functions F and G. Find the distribution function of XY.

Knowledge used. `durrett_ch2_joint_law_product_measure`; `durrett_ch2_expectation_factorization`; `durrett_ch2_convolution_cdf`.

Solution. Let $H(t) = \mathbb{P}(XY \le t)$. Since $X, Y \ge 0$, $H(t) = 0$ for $t < 0$. For $t \ge 0$, condition on the value of Y and use independence:

$$H(t) = \int_{[0,\infty)} \mathbb{P}(Xy \le t)\, G(dy).$$

If $y = 0$, then $Xy = 0 \le t$, so the inner probability is 1. If $y > 0$, it is $\mathbb{P}(X \le t/y) = F(t/y)$. Thus

$$H(t) = G(0) + \int_{(0,\infty)} F(t/y)\, G(dy), \qquad t \ge 0.$$

Equivalently, with the convention $F(\infty) = 1$, this can be written as

$$H(t) = \int_{[0,\infty)} F(t/y)\, G(dy), \qquad t \ge 0.$$

Exercise 2.1.15

Question. If we want an infinite sequence of coin tossings, we do not have to use Kolmogorov's theorem. Let Ω be the unit interval $(0,1)$ equipped with the Borel sets $\mathcal{F}$ and Lebesgue measure $\mathbb{P}$. Let

$$Y_n(\omega) = \begin{cases} 1, & \lfloor 2^n\omega \rfloor \text{ is odd,} \\ 0, & \lfloor 2^n\omega \rfloor \text{ is even.} \end{cases}$$

Show that $Y_1, Y_2, \dots$ are independent with $\mathbb{P}(Y_k = 0) = \mathbb{P}(Y_k = 1) = 1/2$.

Knowledge used. durrett_ch2_construct_independent_variables; durrett_ch2_event_independence; durrett_ch2_pi_lambda_independence.

Solution. Except on the countable set of dyadic rationals, every $\omega \in (0,1)$ has a unique binary expansion

$$\omega = .b_1b_2b_3\cdots, \qquad b_i \in \{0,1\}.$$

The parity of $\lfloor 2^n\omega \rfloor$ is exactly the nth binary digit b_n. The exceptional dyadic rationals have Lebesgue measure zero, so they do not affect the probabilities below.

Fix n and a binary vector $(\varepsilon_1, \dots, \varepsilon_n)$. The set

$$\{\omega : Y_1 = \varepsilon_1, \dots, Y_n = \varepsilon_n\}$$

is, up to endpoints, one dyadic interval of length 2^{-n}: it is the set of numbers whose first n binary digits are the prescribed vector. Therefore

$$\mathbb{P}(Y_1 = \varepsilon_1, \dots, Y_n = \varepsilon_n) = 2^{-n} = \prod_{k=1}^{n} \frac{1}{2}.$$

Taking $n = 1$ gives $\mathbb{P}(Y_k = 0) = \mathbb{P}(Y_k = 1) = 1/2$ for each k, and the finite-dimensional product formula proves that $Y_1, Y_2, \dots$ are independent.

Exercise 2.2.1

Question. Let $X_1, X_2, \dots$ be uncorrelated with $\mathbb{E}X_i = \mu_i$ and $\mathrm{var}(X_i)/i \to 0$ as $i \to \infty$. Let $S_n = X_1 + \cdots + X_n$ and $\nu_n = \mathbb{E}S_n/n$. Then, as $n \to \infty$,

$$S_n/n - \nu_n \to 0$$

in L^2 and in probability.

Knowledge used. durrett_ch2_weak_law_l2; durrett_ch2_iid_finite_variance_wlln; durrett_ch2_poissonization_and_wlln_examples.

Solution. Since the variables are uncorrelated,

$$\mathrm{var}(S_n) = \sum_{i=1}^{n} \mathrm{var}(X_i).$$

Thus

$$\mathbb{E}(S_n/n - \nu_n)^2 = \mathrm{var}(S_n/n) = \frac{1}{n^2} \sum_{i=1}^{n} \mathrm{var}(X_i).$$

Write $a_i = \mathrm{var}(X_i)/i$. Then $a_i \to 0$ and

$$\frac{1}{n^2}\sum_{i=1}^{n}\mathrm{var}(X_i) = \frac{1}{n^2}\sum_{i=1}^{n} i a_i.$$

Given $\varepsilon > 0$, choose N so that $a_i \le \varepsilon$ for $i \ge N$. Then

$$\frac{1}{n^2}\sum_{i=1}^{n} i a_i \le \frac{1}{n^2}\sum_{i=1}^{N-1} i a_i + \varepsilon \frac{1}{n^2}\sum_{i=N}^{n} i.$$

The first term tends to 0, and the second is at most $\varepsilon/2 + o(1)$. Since ε is arbitrary,

$$\mathbb{E}(S_n/n - \nu_n)^2 \to 0.$$

This proves convergence in L^2. Convergence in probability follows from Chebyshev's inequality.

Exercise 2.2.2

Question. The L^2 weak law generalizes immediately to certain dependent sequences. Suppose $\mathbb{E}X_n = 0$ and

$$\mathbb{E}X_n X_m \le r(n-m), \qquad m \le n,$$

with $r(k) \to 0$ as $k \to \infty$. Show that

$$(X_1 + \cdots + X_n)/n \to 0$$

in probability.

Knowledge used. `durrett_ch2_weak_law_l2`; `durrett_ch2_poissonization_and_wlln_examples`.

Solution. Let $S_n = X_1 + \cdots + X_n$. Since $\mathbb{E}X_i = 0$,

$$\mathbb{E}(S_n/n)^2 = \frac{1}{n^2}\mathbb{E}S_n^2.$$

Using the assumed covariance bound,

$$\begin{aligned}\mathbb{E}S_n^2 &= \sum_{i=1}^{n}\mathbb{E}X_i^2 + 2\sum_{1\le m<i\le n}\mathbb{E}X_i X_m \\ &\le n r(0) + 2\sum_{k=1}^{n-1}(n-k)r(k).\end{aligned}$$

Fix $\varepsilon > 0$. Choose K so that $r(k) \le \varepsilon$ for $k \ge K$. Then

$$\frac{1}{n^2}\mathbb{E}S_n^2 \le \frac{r(0)}{n} + \frac{2}{n^2}\sum_{k=1}^{K-1}(n-k)r(k) + \frac{2\varepsilon}{n^2}\sum_{k=K}^{n-1}(n-k).$$

The first two terms tend to 0, while the last term is at most $\varepsilon + o(1)$. Since $\mathbb{E}(S_n/n)^2 \ge 0$ and ε is arbitrary, $\mathbb{E}(S_n/n)^2 \to 0$. Chebyshev's inequality gives $S_n/n \to 0$ in probability.

Exercise 2.2.3

Question. Monte Carlo integration.

(i) Let f be a measurable function on $[0,1]$ with $\int_0^1 |f(x)|\,dx < \infty$. Let $U_1, U_2, \ldots$ be independent and uniformly distributed on $[0,1]$, and let

$$I_n = n^{-1}(f(U_1) + \cdots + f(U_n)).$$

Show that $I_n \to I \equiv \int_0^1 f\,dx$ in probability.

(ii) Suppose $\int_0^1 |f(x)|^2\,dx < \infty$. Use Chebyshev's inequality to estimate $\mathbb{P}(|I_n - I| > a/n^{1/2})$.

Knowledge used. durrett_ch2_iid_finite_mean_wlln; durrett_ch2_iid_finite_variance_wlln; durrett_ch2_weak_law_l2.

Solution. Let $Y_i = f(U_i)$. The Y_i are i.i.d. and

$$\mathbb{E}|Y_1| = \int_0^1 |f(x)|\,dx < \infty, \qquad \mathbb{E}Y_1 = \int_0^1 f(x)\,dx = I.$$

By the finite-mean weak law of large numbers,

$$I_n = \frac{1}{n}\sum_{i=1}^n Y_i \to I$$

in probability.

For (ii), assume $f \in L^2[0,1]$. Then

$$\sigma^2 = \operatorname{var}(f(U_1)) = \int_0^1 f(x)^2\,dx - I^2 < \infty.$$

Since $\operatorname{var}(I_n) = \sigma^2/n$, Chebyshev's inequality gives

$$\mathbb{P}\left(|I_n - I| > \frac{a}{\sqrt{n}}\right) \le \frac{\sigma^2/n}{a^2/n} = \frac{\sigma^2}{a^2}.$$

In particular, for a fixed error threshold $a > 0$ one also has $\mathbb{P}(|I_n - I| > a) \le \sigma^2/(na^2)$.

Exercise 2.2.4

Question. Let $X_1, X_2, \ldots$ be i.i.d. with

$$\mathbb{P}(X_i = (-1)^k k) = \frac{C}{k^2 \log k}, \qquad k \ge 2,$$

where C is chosen to make the sum of the probabilities equal to 1. Show that $\mathbb{E}|X_i| = \infty$, but there is a finite constant μ so that $S_n/n \to \mu$ in probability.

Knowledge used. durrett_ch2_truncation_method; durrett_ch2_iid_finite_mean_wlln; durrett_ch2_infinite_mean_behavior.

Solution. First,

$$\mathbb{E}|X_1| = C\sum_{k=2}^\infty \frac{k}{k^2 \log k} = C\sum_{k=2}^\infty \frac{1}{k \log k} = \infty.$$

However,

$$\mathbb{P}(|X_1| > x) = C \sum_{k>x} \frac{1}{k^2 \log k} \le \frac{C'}{x \log x}$$

for large x, so $x\mathbb{P}(|X_1| > x) \to 0$.

By the truncation weak law, with

$$\mu_n = \mathbb{E}[X_1 \mathbf{1}_{\{|X_1| \le n\}}],$$

we have

$$S_n/n - \mu_n \to 0$$

in probability. It remains only to identify the limit of μ_n. Since

$$\mu_n = C \sum_{2 \le k \le n} \frac{(-1)^k}{k \log k},$$

and the terms $1/(k \log k)$ decrease to 0, the alternating series test implies that μ_n converges to a finite constant

$$\mu = C \sum_{k=2}^{\infty} \frac{(-1)^k}{k \log k}.$$

Therefore $S_n/n \to \mu$ in probability.

Exercise 2.2.5

Question. Let $X_1, X_2, \ldots$ be i.i.d. with

$$\mathbb{P}(X_i > x) = \frac{e}{x \log x}, \qquad x \ge e.$$

Show that $\mathbb{E}|X_i| = \infty$, but there is a sequence of constants $\mu_n \to \infty$ so that

$$S_n/n - \mu_n \to 0$$

in probability.

Knowledge used. `durrett_ch2_truncation_method`; `durrett_ch2_infinite_mean_behavior`; `durrett_ch2_iid_finite_mean_wlln`.

Solution. Since X_i has a positive heavy tail,

$$\mathbb{E}X_1 \ge \int_e^{\infty} \mathbb{P}(X_1 > x)\, dx = \int_e^{\infty} \frac{e}{x \log x}\, dx = \infty.$$

On the other hand,

$$x\mathbb{P}(|X_1| > x) = x\mathbb{P}(X_1 > x) = \frac{e}{\log x} \to 0.$$

The truncation weak law therefore applies. Define

$$\mu_n = \mathbb{E}[X_1 \mathbf{1}_{\{|X_1| \le n\}}].$$

Then

$$S_n/n - \mu_n \to 0$$

in probability. Finally, $\mu_n \to \infty$ by monotone convergence, because $\mathbb{E}X_1 = \infty$.

Exercise 2.2.6

Question.

(i) Show that if $X \geq 0$ is integer valued, then

$$\mathbb{E}X = \sum_{n\geq 1} \mathbb{P}(X \geq n).$$

(ii) Find a similar expression for $\mathbb{E}X^2$.

Knowledge used. durrett_ch1_tail_sum_nonnegative; durrett_ch2_truncation_method.

Solution. If $X = k$, then

$$k = \sum_{n=1}^{\infty} \mathbf{1}_{\{n\leq k\}}.$$

Thus, for integer-valued $X \geq 0$,

$$X = \sum_{n=1}^{\infty} \mathbf{1}_{\{X\geq n\}}.$$

Taking expectations and using Tonelli's theorem gives

$$\mathbb{E}X = \sum_{n=1}^{\infty} \mathbb{P}(X \geq n).$$

Similarly,

$$k^2 = \sum_{n=1}^{k} (2n-1),$$

so

$$X^2 = \sum_{n=1}^{\infty} (2n-1)\mathbf{1}_{\{X\geq n\}}.$$

Therefore

$$\mathbb{E}X^2 = \sum_{n=1}^{\infty} (2n-1)\mathbb{P}(X \geq n).$$

Exercise 2.2.7

Question. Generalize Lemma 2.2.13 to conclude that if

$$H(x) = \int_{(-\infty,x]} h(y)\, dy, \qquad h(y) \geq 0,$$

then

$$\mathbb{E}H(X) = \int_{-\infty}^{\infty} h(y)\mathbb{P}(X \geq y)\, dy.$$

An important special case is $H(x) = \exp(\theta x)$ with $\theta > 0$.

Knowledge used. durrett_ch1_fubini_tonelli; durrett_ch1_tail_sum_nonnegative; durrett_ch2_truncation_method.

Solution. Since $h \geq 0$,

$$H(X) = \int_{-\infty}^{\infty} h(y)\mathbf{1}_{\{y \leq X\}}\, dy.$$

By Tonelli's theorem,

$$\begin{aligned}\mathbb{E}H(X) &= \mathbb{E}\int_{-\infty}^{\infty} h(y)\mathbf{1}_{\{y \leq X\}}\, dy \\ &= \int_{-\infty}^{\infty} h(y)\mathbb{P}(y \leq X)\, dy \\ &= \int_{-\infty}^{\infty} h(y)\mathbb{P}(X \geq y)\, dy.\end{aligned}$$

For $H(x) = e^{\theta x}$ with $\theta > 0$, take $h(y) = \theta e^{\theta y}$. Then

$$\mathbb{E}e^{\theta X} = \int_{-\infty}^{\infty} \theta e^{\theta y}\mathbb{P}(X \geq y)\, dy,$$

with both sides allowed to be $+\infty$.

Exercise 2.2.8

Question. An unfair "fair game." Let

$$p_k = \frac{1}{2^k k(k+1)}, \qquad k = 1, 2, \ldots,$$

and $p_0 = 1 - \sum_{k \geq 1} p_k$. Since

$$\sum_{k=1}^{\infty} 2^k p_k = \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots = 1,$$

if $X_1, X_2, \ldots$ are i.i.d. with

$$\mathbb{P}(X_n = -1) = p_0, \qquad \mathbb{P}(X_n = 2^k - 1) = p_k, \quad k \geq 1,$$

then $\mathbb{E}X_n = 0$. Let $S_n = X_1 + \cdots + X_n$. Use Theorem 2.2.11 with $b_n = 2^{m(n)}$, where

$$m(n) = \min\{m : 2^{-m}m^{-3/2} \leq n^{-1}\},$$

to conclude that

$$S_n/(n/\log_2 n) \to -1$$

in probability.

Knowledge used. durrett_ch2_triangular_array_wlln; durrett_ch2_truncation_method; durrett_ch2_infinite_mean_behavior.

Solution. Let $b_n = 2^{m(n)}$ and truncate X_k at level b_n:

$$\bar{X}_{n,k} = X_k \mathbf{1}_{\{|X_k| \leq b_n\}}.$$

We verify the two hypotheses of Theorem 2.2.11.

First,

$$n\mathbb{P}(|X_1| > b_n) = n \sum_{j\ge m(n)+1} \frac{1}{2^j j(j+1)} \le Cn\, 2^{-m(n)} m(n)^{-2}.$$

By the definition of $m(n)$, $n2^{-m(n)}m(n)^{-3/2} \le 1$, so the last expression is at most $Cm(n)^{-1/2} \to 0$.

Second,

$$\mathbb{E}\bar{X}_{n,1}^2 \le 1 + \sum_{j=1}^{m(n)} \frac{(2^j-1)^2}{2^j j(j+1)} \le C \sum_{j=1}^{m(n)} \frac{2^j}{j(j+1)} \le C\frac{2^{m(n)}}{m(n)^2}.$$

Hence

$$b_n^{-2} \sum_{k=1}^{n} \mathbb{E}\bar{X}_{n,k}^2 \le Cn\, 2^{-m(n)} m(n)^{-2} \le Cm(n)^{-1/2} \to 0.$$

The centering in Theorem 2.2.11 is

$$a_n = n\mathbb{E}\bar{X}_{n,1}.$$

Since $\mathbb{E}X_1 = 0$,

$$\begin{aligned}
\mathbb{E}\bar{X}_{n,1} &= -\sum_{j>m(n)} (2^j-1)p_j \\
&= -\sum_{j>m(n)} \frac{1-2^{-j}}{j(j+1)} \sim -\frac{1}{m(n)}.
\end{aligned}$$

Also the defining minimality of $m(n)$ implies $m(n)/\log_2 n \to 1$ and

$$\frac{b_n}{n/\log_2 n} = \frac{2^{m(n)}\log_2 n}{n} \to 0.$$

Therefore Theorem 2.2.11 gives

$$\frac{S_n - a_n}{n/\log_2 n} \to 0$$

in probability, while

$$\frac{a_n}{n/\log_2 n} \sim -\frac{\log_2 n}{m(n)} \to -1.$$

Combining the two limits proves

$$\frac{S_n}{n/\log_2 n} \to -1$$

in probability.

Exercise 2.2.9

Question. Weak law for positive variables. Suppose $X_1, X_2, \dots$ are i.i.d., $\mathbb{P}(0 \le X_i < \infty) = 1$, and $\mathbb{P}(X_i > x) > 0$ for all x. Let

$$\mu(s) = \int_0^s x\, dF(x), \qquad \nu(s) = \frac{\mu(s)}{s(1-F(s))}.$$

It is known that there exist constants a_n so that $S_n/a_n \to 1$ in probability if and only if $\nu(s) \to \infty$ as $s \to \infty$. Pick $b_n \ge 1$ so that $n\mu(b_n) = b_n$ for large n, and use Theorem 2.2.11 to prove that the condition is sufficient.

Knowledge used. durrett_ch2_triangular_array_wlln; durrett_ch2_truncation_method; durrett_ch2_infinite_mean_behavior.

Solution. Assume $\nu(s) \to \infty$. Choose b_n so that

$$n\mu(b_n) = b_n.$$

Since $\mathbb{P}(X_i > x) > 0$ for all x, we have $b_n \to \infty$.

Apply Theorem 2.2.11 to the triangular array $X_{n,k} = X_k$ with truncation level b_n. The centering is

$$a_n = \sum_{k=1}^{n} \mathbb{E}[X_k \mathbf{1}_{\{X_k \le b_n\}}] = n\mu(b_n) = b_n.$$

For condition (i),

$$\sum_{k=1}^{n} \mathbb{P}(X_{n,k} > b_n) = n(1 - F(b_n)) = \frac{b_n(1 - F(b_n))}{\mu(b_n)} = \frac{1}{\nu(b_n)} \to 0.$$

For condition (ii), it is enough to show

$$\frac{\mathbb{E}[X_1^2 \mathbf{1}_{\{X_1 \le b_n\}}]}{b_n \mu(b_n)} \to 0.$$

Using the tail-integral identity,

$$\mathbb{E}[X_1^2 \mathbf{1}_{\{X_1 \le b\}}] \le 2 \int_0^b x\mathbb{P}(X_1 > x)\, dx.$$

Since $\nu(s) \to \infty$, for every $\varepsilon > 0$ there is M such that

$$x\mathbb{P}(X_1 > x) \le \varepsilon\mu(x) \le \varepsilon\mu(b), \qquad M \le x \le b.$$

Thus

$$2 \int_0^b x\mathbb{P}(X_1 > x)\, dx \le C_M + 2\varepsilon b\mu(b),$$

where $C_M = 2\int_0^M x\mathbb{P}(X_1 > x)\, dx$ is fixed. Dividing by $b\mu(b)$ and letting $b \to \infty$ gives a limsup at most 2ε. Since ε is arbitrary, condition (ii) holds.

Theorem 2.2.11 now yields

$$\frac{S_n - b_n}{b_n} \to 0$$

in probability. Hence $S_n/b_n \to 1$ in probability, proving the sufficiency of $\nu(s) \to \infty$.

Exercise 2.3.1

Question. Prove that

$$\mathbb{P}(\limsup A_n) \ge \limsup \mathbb{P}(A_n), \qquad \mathbb{P}(\liminf A_n) \le \liminf \mathbb{P}(A_n).$$

Knowledge used. durrett_ch2_limsup_events; durrett_ch1_measure_continuity.

Solution. Let

$$B_m = \bigcup_{n \ge m} A_n.$$

Then $B_m \downarrow \limsup A_n$, so continuity from above gives

$$\mathbb{P}(\limsup A_n) = \lim_{m\to\infty} \mathbb{P}(B_m).$$

For each m, B_m contains every A_n with $n \geq m$, so

$$\mathbb{P}(B_m) \geq \sup_{n\geq m} \mathbb{P}(A_n).$$

Letting $m \to \infty$ gives

$$\mathbb{P}(\limsup A_n) \geq \lim_{m\to\infty} \sup_{n\geq m} \mathbb{P}(A_n) = \limsup_{n\to\infty} \mathbb{P}(A_n).$$

Similarly, let

$$C_m = \bigcap_{n\geq m} A_n.$$

Then $C_m \uparrow \liminf A_n$, so

$$\mathbb{P}(\liminf A_n) = \lim_{m\to\infty} \mathbb{P}(C_m).$$

Since $C_m \subseteq A_n$ for every $n \geq m$,

$$\mathbb{P}(C_m) \leq \inf_{n\geq m} \mathbb{P}(A_n).$$

Letting $m \to \infty$ gives

$$\mathbb{P}(\liminf A_n) \leq \liminf_{n\to\infty} \mathbb{P}(A_n).$$

Exercise 2.3.2

Question. Prove directly from the definition that if f is continuous and $X_n \to X$ in probability, then $f(X_n) \to f(X)$ in probability.

Knowledge used. `durrett_ch2_poissonization_and_wlln_examples`; `durrett_ch1_composition_measurable`; `durrett_ch1_bounded_convergence`.

Solution. Fix $\varepsilon > 0$ and $\eta > 0$. Since X is finite almost surely, choose M so large that

$$\mathbb{P}(|X| > M) < \eta.$$

The function f is uniformly continuous on the compact interval $[-M-1, M+1]$. Hence there is $0 < \delta < 1$ such that if $x, y \in [-M-1, M+1]$ and $|x-y| < \delta$, then $|f(x) - f(y)| < \varepsilon$.

On the event $\{|X| \leq M,\ |X_n - X| < \delta\}$, we have $X_n \in [-M-1, M+1]$, so $|f(X_n) - f(X)| < \varepsilon$. Therefore

$$\mathbb{P}(|f(X_n) - f(X)| \geq \varepsilon) \leq \mathbb{P}(|X| > M) + \mathbb{P}(|X_n - X| \geq \delta).$$

Taking $\limsup$ and using $X_n \to X$ in probability,

$$\limsup_{n\to\infty} \mathbb{P}(|f(X_n) - f(X)| \geq \varepsilon) \leq \eta.$$

Since $\eta > 0$ is arbitrary, the left-hand side is 0. Thus $f(X_n) \to f(X)$ in probability.

Exercise 2.3.3

Question. Let ℓ_n be the length of the head run at time n. Show that

$$\limsup_{n\to\infty} \frac{\ell_n}{\log_2 n} = 1, \qquad \liminf_{n\to\infty} \ell_n = 0$$

almost surely.

Knowledge used. `durrett_ch2_borel_cantelli_first`; `durrett_ch2_borel_cantelli_second`; `durrett_ch2_bc_independent_trials`.

Solution. Assume the tosses are i.i.d. with $\mathbb{P}(\text{head}) = \mathbb{P}(\text{tail}) = 1/2$. For the upper bound, fix $\varepsilon > 0$. Since

$$\mathbb{P}(\ell_n \geq (1+\varepsilon)\log_2 n) \leq 2^{-(1+\varepsilon)\log_2 n} = n^{-(1+\varepsilon)},$$

the series of these probabilities is finite. By the first Borel-Cantelli lemma,

$$\ell_n < (1+\varepsilon)\log_2 n$$

eventually almost surely. Hence

$$\limsup_{n\to\infty} \frac{\ell_n}{\log_2 n} \leq 1 \qquad \text{a.s.}$$

For the lower bound, fix $\varepsilon \in (0,1)$ and look at the dyadic block of times $\{2^k + 1, \ldots, 2^{k+1}\}$. Split this block into disjoint subblocks of length

$$r_k = \lfloor (1-\varepsilon)k \rfloor.$$

The probability that a given subblock consists entirely of heads is $2^{-r_k} \geq 2^{-(1-\varepsilon)k-1}$. There are at least $c2^k/k$ disjoint subblocks, so the probability that none of them is all heads is bounded by

$$\left(1 - 2^{-r_k}\right)^{c2^k/k} \leq \exp\{-c' 2^{\varepsilon k}/k\}.$$

This is summable in k. By Borel-Cantelli, eventually every dyadic block contains a run of heads of length at least r_k. Therefore

$$\limsup_{n\to\infty} \frac{\ell_n}{\log_2 n} \geq 1 - \varepsilon \qquad \text{a.s.}$$

Letting $\varepsilon \downarrow 0$ gives the lower bound 1.

Finally, $\ell_n = 0$ whenever the nth toss is a tail. The tail events are independent and have probabilities summing to infinity, so by the second Borel-Cantelli lemma tails occur infinitely often. Hence

$$\liminf_{n\to\infty} \ell_n = 0 \qquad \text{a.s.}$$

Exercise 2.3.4

Question. Fatou's lemma. Suppose $X_n \geq 0$ and $X_n \to X$ in probability. Show that

$$\liminf_{n\to\infty} \mathbb{E}X_n \geq \mathbb{E}X.$$

Knowledge used. `durrett_ch1_fatou_integral`; `durrett_ch2_subsequence_bc_upgrade`; `durrett_ch2_limsup_events`.

Solution. Let $a = \liminf_n \mathbb{E}X_n$. If $a = \infty$, there is nothing to prove. Otherwise choose a subsequence n_k such that

$$\mathbb{E}X_{n_k} \to a.$$

Since $X_n \to X$ in probability, the subsequence principle gives a further subsequence $X_{n_{k_j}}$ such that

$$X_{n_{k_j}} \to X \qquad \text{a.s.}$$

By the ordinary Fatou lemma for almost sure convergence and nonnegative random variables,

$$\mathbb{E}X \leq \liminf_{j\to\infty} \mathbb{E}X_{n_{k_j}} = a.$$

Thus $\mathbb{E}X \leq \liminf_n \mathbb{E}X_n$.

Exercise 2.3.5

Question. Dominated convergence. Suppose $X_n \to X$ in probability and either

(a) $|X_n| \leq Y$ with $\mathbb{E}Y < \infty$, or

(b) there is a continuous function g with $g(x) > 0$ for large x and $|x|/g(x) \to 0$ as $|x| \to \infty$ so that $\mathbb{E}g(X_n) \leq C < \infty$ for all n.

Show that $\mathbb{E}X_n \to \mathbb{E}X$.

Knowledge used. `durrett_ch1_dominated_convergence`; `durrett_ch1_bounded_convergence`; `durrett_ch2_subsequence_bc_upgrade`.

Solution. For (a), every subsequence of X_n has a further subsequence $X_{n_{k_j}}$ that converges to X almost surely. Since $|X_{n_{k_j}}| \leq Y$ and Y is integrable, dominated convergence gives

$$\mathbb{E}X_{n_{k_j}} \to \mathbb{E}X.$$

Thus every subsequence of the numerical sequence $\mathbb{E}X_n$ has a further subsequence converging to $\mathbb{E}X$, so $\mathbb{E}X_n \to \mathbb{E}X$.

For (b), add a constant to g if necessary so that $g \geq 1$ everywhere; this only changes the constant C. Since g is continuous and $X_n \to X$ in probability, Exercise 2.3.2 gives $g(X_n) \to g(X)$ in probability. By Exercise 2.3.4,

$$\mathbb{E}g(X) \leq \liminf_n \mathbb{E}g(X_n) \leq C.$$

We now prove uniform integrability of X_n. Given $\varepsilon > 0$, choose K so large that

$$|x| \leq \varepsilon g(x), \qquad |x| > K.$$

Then, for all n,

$$\mathbb{E}\big(|X_n| \mathbf{1}_{\{|X_n|>K\}}\big) \leq \varepsilon \mathbb{E}g(X_n) \leq \varepsilon C,$$

and the same argument gives

$$\mathbb{E}\big(|X| \mathbf{1}_{\{|X|>K\}}\big) \leq \varepsilon C.$$

Let $\phi_K(x) = (-K) \vee (x \wedge K)$. This function is bounded and continuous, so $\mathbb{E}\phi_K(X_n) \to \mathbb{E}\phi_K(X)$. Therefore

$$\limsup_{n\to\infty} |\mathbb{E}X_n - \mathbb{E}X| \leq 2\varepsilon C.$$

Since ε is arbitrary, $\mathbb{E}X_n \to \mathbb{E}X$.

Exercise 2.3.6

Question. Metric for convergence in probability. Show

(a) that
$$d(X,Y) = \mathbb{E}\left(\frac{|X-Y|}{1+|X-Y|}\right)$$
defines a metric on the set of random variables, i.e. $d(X,Y)=0$ if and only if $X=Y$ a.s., $d(X,Y)=d(Y,X)$, and $d(X,Z)\le d(X,Y)+d(Y,Z)$;

(b) that $d(X_n,X)\to 0$ if and only if $X_n\to X$ in probability.

Knowledge used. durrett_ch2_poissonization_and_wlln_examples; durrett_ch1_markov_chebyshev; durrett_ch1_bounded_convergence.

Solution. Let $\varphi(t)=t/(1+t)$ for $t\ge 0$. Then $\varphi(t)\ge 0$ and $\varphi(t)=0$ if and only if $t=0$. Hence $d(X,Y)=0$ if and only if $|X-Y|=0$ a.s., i.e. $X=Y$ a.s. Symmetry is immediate.

For the triangle inequality, note that φ is increasing and subadditive:
$$\varphi(a+b)\le\varphi(a)+\varphi(b),\qquad a,b\ge 0.$$
This follows, for example, from the metric transform proved in Exercise 2.1.3. Since $|X-Z|\le |X-Y|+|Y-Z|$,
$$\varphi(|X-Z|)\le\varphi(|X-Y|)+\varphi(|Y-Z|).$$
Taking expectations gives $d(X,Z)\le d(X,Y)+d(Y,Z)$.

Now suppose $X_n\to X$ in probability. Then $\varphi(|X_n-X|)\to 0$ in probability and $0\le\varphi(|X_n-X|)\le 1$. For any $\eta>0$,
$$\mathbb{E}\varphi(|X_n-X|)\le\eta+\mathbb{P}(\varphi(|X_n-X|)>\eta),$$
and the right side has limsup at most η. Letting $\eta\downarrow 0$ shows $d(X_n,X)\to 0$.

Conversely, if $d(X_n,X)\to 0$, then for every $\varepsilon>0$,
$$\mathbb{P}(|X_n-X|>\varepsilon)=\mathbb{P}\left(\varphi(|X_n-X|)>\frac{\varepsilon}{1+\varepsilon}\right)\le\frac{1+\varepsilon}{\varepsilon}d(X_n,X),$$
which tends to 0. Thus $X_n\to X$ in probability.

Exercise 2.3.7

Question. Show that random variables are a complete space under the metric defined in the previous exercise, i.e. if $d(X_m,X_n)\to 0$ whenever $m,n\to\infty$, then there is a random variable X_∞ so that $X_n\to X_\infty$ in probability.

Knowledge used. durrett_ch2_borel_cantelli_first; durrett_ch2_limsup_events; durrett_ch2_subsequence_bc_upgrade.

Solution. By Exercise 2.3.6, d-Cauchy is the same as Cauchy in probability. Choose a subsequence n_k so that
$$\mathbb{P}(|X_{n_{k+1}}-X_{n_k}|>2^{-k})\le 2^{-k}.$$
By Borel-Cantelli, with probability one,
$$|X_{n_{k+1}}-X_{n_k}|\le 2^{-k}$$

for all sufficiently large k. Hence X_{n_k} is almost surely a Cauchy sequence in $\mathbb{R}$ and therefore converges almost surely to some finite random variable X_∞.

It remains to show that the whole sequence converges to X_∞ in probability. Fix $\varepsilon > 0$ and $\eta > 0$. Since X_n is Cauchy in probability, choose K so large that for all $m, n \geq n_K$,

$$\mathbb{P}(|X_m - X_n| > \varepsilon/2) < \eta.$$

Also choose K larger if necessary so that

$$\mathbb{P}(|X_{n_K} - X_\infty| > \varepsilon/2) < \eta.$$

Then for all $n \geq n_K$,

$$\mathbb{P}(|X_n - X_\infty| > \varepsilon) \leq \mathbb{P}(|X_n - X_{n_K}| > \varepsilon/2) + \mathbb{P}(|X_{n_K} - X_\infty| > \varepsilon/2) < 2\eta.$$

Since η is arbitrary, $X_n \to X_\infty$ in probability.

Exercise 2.3.8

Question. Let A_n be a sequence of independent events with $\mathbb{P}(A_n) < 1$ for all n. Show that

$$\mathbb{P}\left(\bigcup_n A_n\right) = 1$$

implies $\sum_n \mathbb{P}(A_n) = \infty$ and hence

$$\mathbb{P}(A_n \text{ i.o.}) = 1.$$

Knowledge used. `durrett_ch2_borel_cantelli_second`; `durrett_ch2_bc_independent_trials`; `durrett_ch2_event_independence`.

Solution. Suppose, toward a contradiction, that

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty.$$

Since $\mathbb{P}(A_n) < 1$ for every n and $\mathbb{P}(A_n) \to 0$, the infinite product

$$\prod_{n=1}^{\infty} (1 - \mathbb{P}(A_n))$$

is strictly positive. Indeed, after finitely many terms we have $\mathbb{P}(A_n) \leq 1/2$, and then

$$\log(1 - \mathbb{P}(A_n)) \geq -2\mathbb{P}(A_n),$$

so the tail product is positive.

By independence,

$$\mathbb{P}\left(\bigcap_{n=1}^{\infty} A_n^c\right) = \prod_{n=1}^{\infty} (1 - \mathbb{P}(A_n)) > 0.$$

This contradicts $\mathbb{P}(\cup_n A_n) = 1$. Hence $\sum_n \mathbb{P}(A_n) = \infty$. The second Borel-Cantelli lemma then gives

$$\mathbb{P}(A_n \text{ i.o.}) = 1.$$

Exercise 2.3.9

Question.

(i) If $\mathbb{P}(A_n) \to 0$ and
$$\sum_{n=1}^{\infty} \mathbb{P}(A_n^c \cap A_{n+1}) < \infty,$$
then $\mathbb{P}(A_n \text{ i.o.}) = 0$.

(ii) Find an example of a sequence A_n to which the result in (i) can be applied but the Borel-Cantelli lemma cannot.

Knowledge used. durrett_ch2_borel_cantelli_first; durrett_ch2_limsup_events; durrett_ch2_bc_tail_bounds.

Solution. Let $B_n = A_n^c \cap A_{n+1}$. If A_n occurs infinitely often, then either A_n occurs eventually always, or there are infinitely many transitions from A_n^c to A_{n+1}. Thus
$$\limsup A_n \subseteq \liminf A_n \ \cup \ \limsup B_n.$$
By Exercise 2.3.1,
$$\mathbb{P}(\liminf A_n) \le \liminf_n \mathbb{P}(A_n) = 0.$$
Also, since $\sum_n \mathbb{P}(B_n) < \infty$, the first Borel-Cantelli lemma gives
$$\mathbb{P}(\limsup B_n) = 0.$$
Therefore $\mathbb{P}(A_n \text{ i.o.}) = 0$.

For (ii), take $\Omega = (0,1)$ with Lebesgue measure and
$$A_n = (0, 1/n).$$
Then $\mathbb{P}(A_n) = 1/n$, so $\sum_n \mathbb{P}(A_n) = \infty$ and the ordinary Borel-Cantelli lemma cannot be applied. However $A_{n+1} \subseteq A_n$, so
$$A_n^c \cap A_{n+1} = \varnothing$$
for every n, and $\mathbb{P}(A_n) \to 0$. Part (i) applies and gives $\mathbb{P}(A_n \text{ i.o.}) = 0$.

Exercise 2.3.10

Question. Kochen-Stone lemma. Suppose $\sum \mathbb{P}(A_k) = \infty$. Use Exercises 1.6.6 and 2.3.1 to show that if
$$\limsup_{n\to\infty} \frac{\left(\sum_{k=1}^{n} \mathbb{P}(A_k)\right)^2}{\sum_{1\le j,k\le n} \mathbb{P}(A_j \cap A_k)} = \alpha > 0,$$
then
$$\mathbb{P}(A_n \text{ i.o.}) \ge \alpha.$$
The case $\alpha = 1$ contains Theorem 2.3.7.

Knowledge used. durrett_ch2_limsup_events; durrett_ch2_borel_cantelli_second; durrett_ch2_indicator_independence; durrett_ch1_markov_chebyshev.

Solution. Let

$$S_n = \sum_{k=1}^n \mathbf{1}_{A_k}, \qquad m_n = \mathbb{E}S_n = \sum_{k=1}^n \mathbb{P}(A_k).$$

Then $m_n \to \infty$ and

$$\mathbb{E}S_n^2 = \sum_{1\le j,k\le n} \mathbb{P}(A_j \cap A_k).$$

Exercise 1.6.6, applied to the nonnegative random variable S_n, gives the second-moment lower bound

$$\mathbb{P}(S_n > 0) \ge \frac{(\mathbb{E}S_n)^2}{\mathbb{E}S_n^2} = \frac{m_n^2}{\sum_{1\le j,k\le n} \mathbb{P}(A_j \cap A_k)}.$$

Thus

$$\mathbb{P}\left(\bigcup_{k=1}^n A_k\right) \ge \frac{m_n^2}{\sum_{1\le j,k\le n} \mathbb{P}(A_j \cap A_k)}.$$

We need the same estimate for tails. Fix $r \ge 1$ and define

$$S_{r,n} = \sum_{k=r}^n \mathbf{1}_{A_k}, \qquad m_{r,n} = \sum_{k=r}^n \mathbb{P}(A_k).$$

The same second-moment argument gives

$$\mathbb{P}\left(\bigcup_{k=r}^n A_k\right) \ge \frac{m_{r,n}^2}{\sum_{r\le j,k\le n} \mathbb{P}(A_j \cap A_k)}.$$

Along a subsequence for which the original ratio tends to α, the finite number of omitted initial terms and cross terms are lower order: the numerator changes by $o(m_n^2)$, and the denominator changes by at most $O(m_n)$ plus a constant. Hence

$$\limsup_{n\to\infty} \frac{m_{r,n}^2}{\sum_{r\le j,k\le n} \mathbb{P}(A_j \cap A_k)} \ge \alpha.$$

Therefore

$$\mathbb{P}\left(\bigcup_{k=r}^\infty A_k\right) \ge \alpha \qquad \text{for every } r.$$

Finally,

$$\limsup A_n = \bigcap_{r=1}^\infty \bigcup_{k=r}^\infty A_k,$$

and the sets $\cup_{k=r}^\infty A_k$ decrease as r increases. By continuity from above,

$$\mathbb{P}(A_n \text{ i.o.}) = \lim_{r\to\infty} \mathbb{P}\left(\bigcup_{k=r}^\infty A_k\right) \ge \alpha.$$

Exercise 2.3.11

Question. Let $X_1, X_2, \ldots$ be independent with $\mathbb{P}(X_n = 1) = p_n$ and $\mathbb{P}(X_n = 0) = 1 - p_n$. Show that

(i) $X_n \to 0$ in probability if and only if $p_n \to 0$;

(ii) $X_n \to 0$ almost surely if and only if $\sum_n p_n < \infty$.

Knowledge used. `durrett_ch2_borel_cantelli_first`; `durrett_ch2_borel_cantelli_second`; `durrett_ch2_limsup_events`.

Solution. For $0 < \varepsilon < 1$,

$$\mathbb{P}(|X_n| > \varepsilon) = \mathbb{P}(X_n = 1) = p_n.$$

Thus $X_n \to 0$ in probability if and only if $p_n \to 0$.

For almost sure convergence, observe that $X_n \to 0$ a.s. if and only if the events $\{X_n = 1\}$ occur only finitely often. If $\sum_n p_n < \infty$, the first Borel-Cantelli lemma gives

$$\mathbb{P}(X_n = 1 \text{ i.o.}) = 0.$$

Conversely, if $\sum_n p_n = \infty$, then the events $\{X_n = 1\}$ are independent, so the second Borel-Cantelli lemma gives

$$\mathbb{P}(X_n = 1 \text{ i.o.}) = 1.$$

Hence $X_n \not\to 0$ a.s. in that case.

Exercise 2.3.12

Question. Let $X_1, X_2, \ldots$ be a sequence of random variables on $(\Omega, \mathcal{F}, \mathbb{P})$, where Ω is a countable set and $\mathcal{F}$ consists of all subsets of Ω. Show that $X_n \to X$ in probability implies $X_n \to X$ almost surely.

Knowledge used. `durrett_ch2_limsup_events`; `durrett_ch2_bc_tail_bounds`.

Solution. Let

$$\Omega_+ = \{\omega \in \Omega : \mathbb{P}(\{\omega\}) > 0\}.$$

Since Ω is countable, $\mathbb{P}(\Omega_+) = 1$. Fix $\omega \in \Omega_+$. We claim that $X_n(\omega) \to X(\omega)$.

If not, then for some $\varepsilon > 0$ there is a subsequence n_k such that

$$|X_{n_k}(\omega) - X(\omega)| > \varepsilon \qquad \text{for all } k.$$

Hence

$$\mathbb{P}(|X_{n_k} - X| > \varepsilon) \geq \mathbb{P}(\{\omega\}) > 0$$

for all k, contradicting convergence in probability. Therefore $X_n(\omega) \to X(\omega)$ for every $\omega \in \Omega_+$, and so $X_n \to X$ almost surely.

Exercise 2.3.13

Question. If X_n is any sequence of random variables, there are constants $c_n \to \infty$ so that

$$X_n/c_n \to 0$$

almost surely.

Knowledge used. durrett_ch2_borel_cantelli_first; durrett_ch2_bc_tail_bounds.

Solution. For each n, choose $c_n \geq n$ so large that

$$\mathbb{P}(|X_n| > c_n/j) \leq 2^{-n}, \qquad j = 1, \ldots, n.$$

This is possible because $\mathbb{P}(|X_n| > t) \to 0$ as $t \to \infty$ for each fixed n. Clearly $c_n \to \infty$.

Fix $\varepsilon > 0$ and choose an integer j with $1/j < \varepsilon$. For all $n \geq j$,

$$\mathbb{P}(|X_n| > \varepsilon c_n) \leq \mathbb{P}(|X_n| > c_n/j) \leq 2^{-n}.$$

Thus

$$\sum_{n=1}^{\infty} \mathbb{P}(|X_n| > \varepsilon c_n) < \infty.$$

By Borel-Cantelli, $|X_n| > \varepsilon c_n$ only finitely often almost surely. Applying this to $\varepsilon = 1/m$, $m = 1, 2, \ldots$, gives $X_n/c_n \to 0$ almost surely.

Exercise 2.3.14

Question. Let $X_1, X_2, \ldots$ be independent. Show that

$$\sup_n X_n < \infty \quad \text{a.s.}$$

if and only if

$$\sum_n \mathbb{P}(X_n > A) < \infty$$

for some A.

Knowledge used. durrett_ch2_borel_cantelli_first; durrett_ch2_borel_cantelli_second; durrett_ch2_bc_tail_bounds.

Solution. If $\sum_n \mathbb{P}(X_n > A) < \infty$, then by the first Borel-Cantelli lemma, $X_n > A$ only finitely often a.s. Thus all sufficiently large n have $X_n \leq A$. The maximum of the remaining finitely many finite random variables is finite a.s., so $\sup_n X_n < \infty$ a.s.

Conversely, suppose $\sup_n X_n < \infty$ a.s. If

$$\sum_n \mathbb{P}(X_n > A) = \infty$$

for every integer A, then, by independence and the second Borel-Cantelli lemma,

$$\mathbb{P}(X_n > A \text{ i.o.}) = 1 \qquad \text{for every integer } A.$$

Intersecting over $A = 1, 2, \ldots$ gives $\sup_n X_n = \infty$ a.s., a contradiction. Hence for some integer A the series is finite.

Exercise 2.3.15

Question. Let $X_1, X_2, \dots$ be i.i.d. with

$$\mathbb{P}(X_i > x) = e^{-x},$$

and let $M_n = \max_{1 \le m \le n} X_m$. Show that

(i) $\limsup_{n\to\infty} X_n / \log n = 1$ a.s.;

(ii) $M_n / \log n \to 1$ a.s.

Knowledge used. durrett_ch2_borel_cantelli_first; durrett_ch2_borel_cantelli_second; durrett_ch2_bc_independent_trials.

Solution. Fix $\varepsilon > 0$. Since

$$\mathbb{P}(X_n > (1+\varepsilon)\log n) = n^{-(1+\varepsilon)},$$

the first Borel-Cantelli lemma implies

$$X_n \le (1+\varepsilon)\log n$$

eventually a.s. Hence

$$\limsup_{n\to\infty} \frac{X_n}{\log n} \le 1+\varepsilon \qquad \text{a.s.}$$

Letting $\varepsilon \downarrow 0$ gives the upper bound.

For the lower bound,

$$\mathbb{P}(X_n > (1-\varepsilon)\log n) = n^{-(1-\varepsilon)},$$

and the series over n diverges. The events are independent, so by the second Borel-Cantelli lemma they occur infinitely often. Hence

$$\limsup_{n\to\infty} \frac{X_n}{\log n} \ge 1-\varepsilon \qquad \text{a.s.}$$

Letting $\varepsilon \downarrow 0$ proves (i).

For (ii), the upper bound in (i) implies that eventually $X_n \le (1+\varepsilon)\log n$, and hence

$$M_n \le C(\omega) \vee (1+\varepsilon)\log n,$$

so $\limsup M_n/\log n \le 1+\varepsilon$ a.s. For the lower bound,

$$\mathbb{P}(M_n \le (1-\varepsilon)\log n) = \left(1 - n^{-(1-\varepsilon)}\right)^n \le \exp(-n^{\varepsilon}).$$

The last bound is summable, so by Borel-Cantelli $M_n > (1-\varepsilon)\log n$ eventually a.s. Letting $\varepsilon \downarrow 0$ gives $M_n/\log n \to 1$ a.s.

Exercise 2.3.16

Question. Let $X_1, X_2, \dots$ be i.i.d. with distribution F, let $\lambda_n \uparrow \infty$, and let

$$A_n = \left\{ \max_{1 \le m \le n} X_m > \lambda_n \right\}.$$

Show that $\mathbb{P}(A_n \text{ i.o.}) = 0$ or 1 according as

$$\sum_{n \ge 1} (1 - F(\lambda_n)) < \infty \quad \text{or} \quad = \infty.$$

Knowledge used. durrett_ch2_borel_cantelli_first; durrett_ch2_borel_cantelli_second; durrett_ch2_bc_tail_bounds.

Solution. Let $B_n = \{X_n > \lambda_n\}$. Then

$$\mathbb{P}(B_n) = 1 - F(\lambda_n).$$

If $\sum_n \mathbb{P}(B_n) < \infty$, then B_n occurs only finitely often a.s. If A_n occurs infinitely often, then infinitely often some $m \le n$ has $X_m > \lambda_n$. Since $\lambda_n \to \infty$, a fixed X_m can only be responsible for finitely many such n. Therefore A_n i.o. would force B_m i.o., which has probability 0. Hence $\mathbb{P}(A_n \text{ i.o.}) = 0$.

If $\sum_n \mathbb{P}(B_n) = \infty$, then the independent events B_n occur infinitely often by the second Borel-Cantelli lemma. Since $B_n \subseteq A_n$, it follows that A_n occurs infinitely often with probability 1.

Exercise 2.3.17

Question. Let $Y_1, Y_2, \dots$ be i.i.d. Find necessary and sufficient conditions for

(i) $Y_n/n \to 0$ almost surely;

(ii) $(\max_{m \le n} Y_m)/n \to 0$ almost surely;

(iii) $(\max_{m \le n} Y_m)/n \to 0$ in probability;

(iv) $Y_n/n \to 0$ in probability.

Knowledge used. durrett_ch2_borel_cantelli_first; durrett_ch2_borel_cantelli_second; durrett_ch2_max_negligible_condition.

Solution. For (i), $Y_n/n \to 0$ a.s. if and only if for every $\varepsilon > 0$,

$$\mathbb{P}(|Y_n| > \varepsilon n \text{ i.o.}) = 0.$$

By Borel-Cantelli and independence, this is equivalent to

$$\sum_{n=1}^{\infty} \mathbb{P}(|Y_1| > \varepsilon n) < \infty \qquad \text{for every } \varepsilon > 0.$$

By the tail-sum criterion, this is equivalent to

$$\mathbb{E}|Y_1| < \infty.$$

For (ii), since $\max_{m \le n} Y_m \ge Y_1$ and $Y_1/n \to 0$, only the upper tail matters. The condition is

$$\sum_{n=1}^{\infty} \mathbb{P}(Y_1 > \varepsilon n) < \infty \qquad \text{for every } \varepsilon > 0,$$

which is equivalent to

$$\mathbb{E}(Y_1^+) < \infty.$$

For (iii),

$$\mathbb{P}\left(\max_{m \le n} Y_m > \varepsilon n\right) = 1 - (1 - \mathbb{P}(Y_1 > \varepsilon n))^n .$$

This tends to 0 if and only if

$$n\mathbb{P}(Y_1 > \varepsilon n) \to 0 \qquad \text{for every } \varepsilon > 0,$$

equivalently

$$x\mathbb{P}(Y_1 > x) \to 0 \qquad (x \to \infty).$$

For (iv), for every $\varepsilon > 0$,

$$\mathbb{P}(|Y_n| > \varepsilon n) = \mathbb{P}(|Y_1| > \varepsilon n) \to 0,$$

provided Y_1 is finite a.s. Thus no moment condition is needed beyond finiteness of the random variable.

Exercise 2.3.18

Question. Let $0 \le X_1 \le X_2 \le \cdots$ be random variables with

$$\mathbb{E}X_n \sim an^{\alpha}, \qquad a, \alpha > 0,$$

and

$$\mathrm{var}(X_n) \le Bn^{\beta}, \qquad \beta < 2\alpha.$$

Show that

$$X_n/n^{\alpha} \to a$$

almost surely.

Knowledge used. `durrett_ch2_subsequence_bc_upgrade`; `durrett_ch1_markov_chebyshev`; `durrett_ch2_slln_normalized_sums`.

Solution. Choose $r > 0$ so large that

$$r(2\alpha - \beta) > 1,$$

and let $n_k = \lfloor k^r \rfloor$. By Chebyshev's inequality,

$$\begin{aligned}\mathbb{P}\left(\left|\frac{X_{n_k} - \mathbb{E}X_{n_k}}{n_k^{\alpha}}\right| > \varepsilon\right) &\le \frac{\mathrm{var}(X_{n_k})}{\varepsilon^2 n_k^{2\alpha}} \\ &\le Ck^{-r(2\alpha-\beta)}.\end{aligned}$$

The last series is summable. By Borel-Cantelli,

$$\frac{X_{n_k} - \mathbb{E}X_{n_k}}{n_k^\alpha} \to 0 \qquad \text{a.s.}$$

Since $\mathbb{E}X_n/n^\alpha \to a$,

$$X_{n_k}/n_k^\alpha \to a \qquad \text{a.s.}$$

Now let $n_k \le n < n_{k+1}$. By monotonicity,

$$\frac{X_{n_k}}{n_{k+1}^\alpha} \le \frac{X_n}{n^\alpha} \le \frac{X_{n_{k+1}}}{n_k^\alpha}.$$

Because $n_{k+1}/n_k \to 1$, both outside terms converge to a a.s. Therefore $X_n/n^\alpha \to a$ a.s.

Exercise 2.3.19

Question. Let X_n be independent Poisson random variables with

$$\mathbb{E}X_n = \lambda_n,$$

and let $S_n = X_1 + \cdots + X_n$. Show that if $\sum_n \lambda_n = \infty$, then

$$S_n/\mathbb{E}S_n \to 1$$

almost surely.

Knowledge used. durrett_ch2_kolmogorov_convergence_criterion; durrett_ch2_slln_normalized_sums; durrett_ch2_weak_law_l2.

Solution. Let

$$m_n = \mathbb{E}S_n = \sum_{k=1}^n \lambda_k.$$

Then $m_n \to \infty$. Since the X_k are independent Poisson random variables,

$$\operatorname{var}(S_n) = \sum_{k=1}^n \operatorname{var}(X_k) = \sum_{k=1}^n \lambda_k = m_n.$$

It is enough to prove

$$\frac{S_n - m_n}{m_n} \to 0 \qquad \text{a.s.}$$

For all sufficiently large n, $m_n > 0$. Put

$$Z_n = \frac{X_n - \lambda_n}{m_n}.$$

Then the Z_n are independent, mean zero, and

$$\sum_n \operatorname{var}(Z_n) = \sum_n \frac{\lambda_n}{m_n^2} < \infty.$$

The last series is finite because $\lambda_n = m_n - m_{n-1}$ and, once $m_{n-1} > 0$,

$$\frac{m_n - m_{n-1}}{m_n^2} \le \int_{m_{n-1}}^{m_n} \frac{dx}{x^2}.$$

Thus $\sum_n Z_n$ converges almost surely by the Kolmogorov convergence criterion. Kronecker's lemma with $a_n = m_n$ then gives

$$\frac{1}{m_n}\sum_{k=1}^{n}(X_k - \lambda_k) \to 0 \qquad \text{a.s.}$$

This is exactly $S_n/m_n \to 1$ a.s.

Exercise 2.3.20

Question. Show that if X_n is the outcome of the nth play of the St. Petersburg game, then

$$\limsup_{n\to\infty} \frac{X_n}{n\log_2 n} = \infty \qquad \text{a.s.}$$

and hence the same result holds for $S_n = X_1 + \cdots + X_n$. This shows that the convergence $S_n/(n\log_2 n) \to 1$ in probability proved in Section 2.2 does not occur a.s.

Knowledge used. `durrett_ch2_borel_cantelli_second`; `durrett_ch2_infinite_mean_behavior`; `durrett_ch2_unfair_fair_game_truncation_scale`.

Solution. In the St. Petersburg game,

$$\mathbb{P}(X_n = 2^j) = 2^{-j}, \qquad j \geq 1.$$

There is a constant $c > 0$ such that for all large x,

$$\mathbb{P}(X_n \geq x) \geq \frac{c}{x}.$$

Fix $M > 0$ and define

$$A_n^{(M)} = \{X_n \geq Mn\log_2 n\}.$$

Then

$$\mathbb{P}(A_n^{(M)}) \geq \frac{c_M}{n\log_2 n}$$

for large n, and the series over n diverges. The events $A_n^{(M)}$ are independent, so by the second Borel-Cantelli lemma they occur infinitely often a.s. Hence

$$\limsup_{n\to\infty} \frac{X_n}{n\log_2 n} \geq M \qquad \text{a.s.}$$

Intersecting over integer $M \geq 1$ gives the limsup equal to infinity. Since $S_n \geq X_n$, the same conclusion holds for $S_n/(n\log_2 n)$.

Exercise 2.4.1

Question. Lazy janitor. Suppose the ith light bulb burns for an amount of time X_i and then remains burned out for time Y_i before being replaced. Suppose the X_i, Y_i are positive and independent with the X's having distribution F and the Y's having distribution G, both of which have finite mean. Let R_t be the amount of time in $[0, t]$ that we have a working light bulb. Show that

$$R_t/t \to \frac{\mathbb{E}X_i}{\mathbb{E}X_i + \mathbb{E}Y_i} \qquad \text{a.s.}$$

Knowledge used. durrett_ch2_iid_finite_mean_slln; durrett_ch2_renewal_process; durrett_ch2_slln_normalized_sums.

Solution. Let $Z_i = X_i + Y_i$ be the length of the ith complete cycle and $T_n = Z_1 + \cdots + Z_n$. Let

$$N(t) = \max\{n : T_n \le t\}.$$

By the strong law,

$$\frac{T_n}{n} \to \mathbb{E}Z_1 = \mathbb{E}X_1 + \mathbb{E}Y_1, \qquad \frac{X_1 + \cdots + X_n}{n} \to \mathbb{E}X_1 \qquad \text{a.s.}$$

Thus $N(t) \to \infty$ and

$$\frac{N(t)}{t} \to \frac{1}{\mathbb{E}X_1 + \mathbb{E}Y_1} \qquad \text{a.s.}$$

The working time up to t differs from $X_1 + \cdots + X_{N(t)}$ by at most the working part of one incomplete cycle. Hence

$$\sum_{i=1}^{N(t)} X_i \le R_t \le \sum_{i=1}^{N(t)+1} X_i.$$

Dividing by t and using the two strong-law limits gives

$$\frac{R_t}{t} \to \mathbb{E}X_1 \cdot \frac{1}{\mathbb{E}X_1 + \mathbb{E}Y_1} = \frac{\mathbb{E}X_1}{\mathbb{E}X_1 + \mathbb{E}Y_1} \qquad \text{a.s.}$$

Exercise 2.4.2

Question. Let $X_0 = (1, 0)$ and define $X_n \in \mathbb{R}^2$ inductively by declaring that X_{n+1} is chosen at random from the ball of radius $|X_n|$ centered at the origin, i.e. $X_{n+1}/|X_n|$ is uniformly distributed on the ball of radius 1 and independent of $X_1, \ldots, X_n$. Prove that

$$n^{-1} \log |X_n| \to c$$

almost surely and compute c.

Knowledge used. durrett_ch2_iid_finite_mean_slln; durrett_ch2_functions_of_independent_blocks; durrett_ch2_slln_normalized_sums.

Solution. Write

$$X_{n+1} = |X_n| U_{n+1},$$

where $U_1, U_2, \ldots$ are i.i.d. uniform random vectors in the unit disk. Then

$$|X_n| = \prod_{k=1}^{n} |U_k|, \qquad \log |X_n| = \sum_{k=1}^{n} \log |U_k|.$$

If $R = |U_1|$, then R has density $2r$ on $0 \le r \le 1$. Hence

$$\mathbb{E}|\log R| = \int_0^1 2r |\log r| \, dr < \infty,$$

and the strong law applies:

$$\frac{1}{n} \log |X_n| \to \mathbb{E} \log R = \int_0^1 2r \log r \, dr = -\frac{1}{2}.$$

Thus $c = -1/2$.

Exercise 2.4.3

Question. Investment problem. At the beginning of each year you can buy bonds for 1 unit that are worth a at the end of the year or stocks that are worth a random amount $V \geq 0$. If you always invest a fixed proportion p of your wealth in bonds, then your wealth at the end of year $n+1$ is

$$W_{n+1} = (ap + (1-p)V_n)W_n.$$

Suppose $V_1, V_2, \ldots$ are i.i.d. with $\mathbb{E}V_n^2 < \infty$ and $\mathbb{E}(V_n^{-2}) < \infty$.

(i) Show that $n^{-1} \log W_n \to c(p)$ a.s.

(ii) Show that $c(p)$ is concave.

(iii) By investigating $c'(0)$ and $c'(1)$, give conditions on V that guarantee that the optimal choice of p is in $(0,1)$.

(iv) Suppose $\mathbb{P}(V=1) = \mathbb{P}(V=4) = 1/2$. Find the optimal p as a function of a.

Knowledge used. durrett_ch2_iid_finite_mean_slln; durrett_ch1_jensen_integral; durrett_ch2_slln_normalized_sums.

Solution. Assume $0 \leq p \leq 1$ and $a > 0$. Iterating the wealth recursion,

$$\log W_n = \log W_0 + \sum_{k=0}^{n-1} \log(ap + (1-p)V_k).$$

The assumptions $\mathbb{E}V^2 < \infty$ and $\mathbb{E}V^{-2} < \infty$ imply that the positive and negative parts of $\log(ap + (1-p)V)$ are integrable for $0 \leq p \leq 1$. Therefore the strong law gives

$$\frac{1}{n} \log W_n \to c(p) := \mathbb{E} \log(ap + (1-p)V) \qquad \text{a.s.}$$

For each fixed $v > 0$, the function

$$p \mapsto \log(ap + (1-p)v)$$

is concave because it is the logarithm of an affine positive function. Taking expectations preserves concavity, so $c(p)$ is concave.

Differentiating under the expectation,

$$c'(p) = \mathbb{E}\left[\frac{a - V}{ap + (1-p)V}\right].$$

Thus

$$c'(0) = a\,\mathbb{E}(1/V) - 1, \qquad c'(1) = 1 - \frac{\mathbb{E}V}{a}.$$

Since c is concave, an interior maximizer is guaranteed when

$$c'(0) > 0 \quad \text{and} \quad c'(1) < 0,$$

that is,

$$a\,\mathbb{E}(1/V) > 1 \quad \text{and} \quad \mathbb{E}V > a.$$

Now suppose $\mathbb{P}(V = 1) = \mathbb{P}(V = 4) = 1/2$. Then

$$c(p) = \frac{1}{2}\log(1 + (a-1)p) + \frac{1}{2}\log(4 + (a-4)p).$$

The endpoint derivative conditions are

$$c'(0) = \frac{5a}{8} - 1, \qquad c'(1) = 1 - \frac{5}{2a}.$$

Hence the optimum is $p = 0$ if $a \le 8/5$, and $p = 1$ if $a \ge 5/2$. For $8/5 < a < 5/2$, solve $c'(p) = 0$:

$$(a-1)(4 + (a-4)p) + (a-4)(1 + (a-1)p) = 0.$$

Thus

$$p^*(a) = \frac{8-5a}{2(a-1)(a-4)}, \qquad 8/5 < a < 5/2.$$

So

$$p^*(a) = \begin{cases} 0, & a \le 8/5, \\ \dfrac{8-5a}{2(a-1)(a-4)}, & 8/5 < a < 5/2, \\ 1, & a \ge 5/2. \end{cases}$$

Exercise 2.5.1

Question. Suppose X_i are i.i.d., $\mathbb{E}X_i = 0$, and $\operatorname{var}(X_i) = C < \infty$. Use Kolmogorov's maximal inequality with $n = m^\alpha$, where $\alpha(2p-1) > 1$, to show that if $S_n = \sum_{i=1}^n X_i$ and $p > 1/2$, then

$$\frac{S_n}{n^p} \to 0 \qquad \text{a.s.}$$

Knowledge used. `durrett_ch2_kolmogorov_maximal_inequality`; `durrett_ch2_borel_cantelli`; `durrett_ch2_subsequence_upgrade`.

Solution. Let $n_m = \lfloor m^\alpha \rfloor$, with $\alpha(2p-1) > 1$. By Kolmogorov's maximal inequality,

$$\mathbb{P}\left(\max_{k \le n_m} |S_k| > \varepsilon n_m^p\right) \le \frac{C n_m}{\varepsilon^2 n_m^{2p}} = \frac{C}{\varepsilon^2} n_m^{1-2p}.$$

Since $n_m^{1-2p} \le c m^{-\alpha(2p-1)}$ and $\alpha(2p-1) > 1$, these probabilities are summable. Borel–Cantelli therefore gives

$$\max_{k \le n_m} \frac{|S_k|}{n_m^p} \to 0 \qquad \text{a.s.}$$

If $n_m < k \le n_{m+1}$, then

$$\frac{|S_k|}{k^p} \le \left(\frac{n_{m+1}}{n_m}\right)^p \max_{j \le n_{m+1}} \frac{|S_j|}{n_{m+1}^p}.$$

Because $n_{m+1}/n_m \to 1$, the right side tends to 0 a.s. Hence $S_n/n^p \to 0$ a.s.

Exercise 2.5.2

Question. Let $p > 0$. Prove the converse of Theorem 2.5.12: if $S_n/n^{1/p} \to 0$ a.s. for i.i.d. X_i, then $\mathbb{E}|X_1|^p < \infty$.
Knowledge used. durrett_ch2_borel_cantelli; durrett_ch2_tail_sum_moment_test; durrett_ch2_slln_normalized_sums.
Solution. Since

$$\frac{X_n}{n^{1/p}} = \frac{S_n}{n^{1/p}} - \left(\frac{n-1}{n}\right)^{1/p} \frac{S_{n-1}}{(n-1)^{1/p}},$$

we have $X_n/n^{1/p} \to 0$ a.s. Hence for every $\varepsilon > 0$,

$$\mathbb{P}(|X_n| > \varepsilon n^{1/p} \text{ i.o.}) = 0.$$

The events are independent, so the second Borel–Cantelli lemma implies

$$\sum_{n=1}^{\infty} \mathbb{P}(|X_1|^p > \varepsilon^p n) < \infty.$$

By the tail-sum criterion this is equivalent to $\mathbb{E}|X_1|^p < \infty$.

Exercise 2.5.3

Question. Let X_i be independent standard normal random variables. Show that, for every fixed t,

$$\sum_{n=1}^{\infty} \frac{X_n \sin(n\pi t)}{n}$$

converges a.s.
Knowledge used. durrett_ch2_two_series_theorem; durrett_ch2_independent_series_variance_test.
Solution. For fixed t, set

$$Y_n = \frac{X_n \sin(n\pi t)}{n}.$$

Then the Y_n are independent, $\mathbb{E}Y_n = 0$, and

$$\sum_{n=1}^{\infty} \operatorname{var}(Y_n) = \sum_{n=1}^{\infty} \frac{\sin^2(n\pi t)}{n^2} \le \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty.$$

The independent-series variance test therefore gives $\sum_n Y_n$ converges a.s.

Exercise 2.5.4

Question. Let X_n be independent with $\mathbb{E}X_n = 0$ and $\operatorname{var}(X_n) = \sigma_n^2$.

(i) If $\sum_n \sigma_n^2/n^2 < \infty$, show that $\sum_n X_n/n$ converges a.s., and hence $n^{-1}\sum_{m=1}^n X_m \to 0$ a.s.

(ii) If $\sum_n \sigma_n^2/n^2 = \infty$ and $\sigma_n^2 \le n^2$, construct independent X_n with $\mathbb{E}X_n = 0$ and $\operatorname{var}(X_n) \le \sigma_n^2$ such that X_n/n, and hence $n^{-1}\sum_{m=1}^n X_m$, does not converge to 0 a.s.

Knowledge used. durrett_ch2_two_series_theorem; durrett_ch2_kronecker_lemma; durrett_ch2_borel_cantelli.

Solution. (i) The variables X_n/n are independent, have mean 0, and

$$\sum_n \mathrm{var}(X_n/n) = \sum_n \frac{\sigma_n^2}{n^2} < \infty.$$

Thus $\sum_n X_n/n$ converges a.s. Kronecker's lemma gives

$$\frac{1}{n}\sum_{m=1}^{n} X_m \to 0 \qquad \text{a.s.}$$

(ii) Let

$$\mathbb{P}(X_n = n) = \mathbb{P}(X_n = -n) = \frac{\sigma_n^2}{2n^2}, \qquad \mathbb{P}(X_n = 0) = 1 - \frac{\sigma_n^2}{n^2}.$$

Then $\mathbb{E}X_n = 0$ and $\mathrm{var}(X_n) = \sigma_n^2$. Also

$$\sum_n \mathbb{P}(|X_n/n| = 1) = \sum_n \frac{\sigma_n^2}{n^2} = \infty.$$

By Borel–Cantelli, $|X_n/n| = 1$ infinitely often a.s.; hence $X_n/n \not\to 0$. If the averages S_n/n converged to 0, then

$$\frac{X_n}{n} = \frac{S_n}{n} - \frac{n-1}{n}\frac{S_{n-1}}{n-1}$$

would also converge to 0, a contradiction.

Exercise 2.5.5

Question. Let $X_n \ge 0$ be independent. Show that the following are equivalent:

(i) $\sum_n X_n < \infty$ a.s.

(ii) $\sum_n \{\mathbb{P}(X_n > 1) + \mathbb{E}[X_n \mathbf{1}_{\{X_n \le 1\}}]\} < \infty$.

(iii) $\sum_n \mathbb{E}[X_n/(1+X_n)] < \infty$.

Knowledge used. durrett_ch2_three_series_theorem; durrett_ch2_borel_cantelli; durrett_ch2_truncation_method.

Solution. For $x \ge 0$,

$$\frac{1}{2}\big(x\mathbf{1}_{\{x\le 1\}} + \mathbf{1}_{\{x>1\}}\big) \le \frac{x}{1+x} \le x\mathbf{1}_{\{x\le 1\}} + \mathbf{1}_{\{x>1\}}.$$

Thus (ii) and (iii) are equivalent.

If (ii) holds, then $\sum_n \mathbb{P}(X_n > 1) < \infty$, so only finitely many large jumps occur a.s. Also

$$\mathbb{E}\sum_n X_n \mathbf{1}_{\{X_n \le 1\}} = \sum_n \mathbb{E}[X_n \mathbf{1}_{\{X_n\le 1\}}] < \infty,$$

so the truncated sum is finite a.s. Hence (i) holds.

Conversely assume (i), and put $Y_n = X_n/(1+X_n)$. Then $0 \le Y_n \le 1$ and $\sum_n Y_n < \infty$ a.s. If $\sum_n \mathbb{E}Y_n = \infty$, then, using $1 - e^{-y} \ge (1-e^{-1})y$ on $0 \le y \le 1$,

$$\mathbb{E}e^{-\sum_n Y_n} = \prod_n \mathbb{E}e^{-Y_n} \le \prod_n \exp\{-(1-e^{-1})\mathbb{E}Y_n\} = 0,$$

which would force $\sum_n Y_n = \infty$ a.s., a contradiction. Therefore $\sum_n \mathbb{E}Y_n < \infty$, which is (iii).

Exercise 2.5.6

Question. Let $\psi(x) = x^2$ for $|x| \le 1$ and $\psi(x) = |x|$ for $|x| \ge 1$. Show that if X_n are independent, $\mathbb{E}X_n = 0$, and $\sum_n \mathbb{E}\psi(X_n) < \infty$, then $\sum_n X_n$ converges a.s.

Knowledge used. `durrett_ch2_three_series_theorem`; `durrett_ch2_truncation_method`.

Solution. Use the three-series theorem with truncation at 1. First,

$$\sum_n \mathbb{P}(|X_n| > 1) \le \sum_n \mathbb{E}[|X_n| \mathbf{1}_{\{|X_n|>1\}}] \le \sum_n \mathbb{E}\psi(X_n) < \infty.$$

For $Y_n = X_n \mathbf{1}_{\{|X_n| \le 1\}}$,

$$\sum_n \operatorname{var}(Y_n) \le \sum_n \mathbb{E}Y_n^2 = \sum_n \mathbb{E}[X_n^2 \mathbf{1}_{\{|X_n| \le 1\}}] < \infty.$$

Finally, since $\mathbb{E}X_n = 0$,

$$|\mathbb{E}Y_n| = |\mathbb{E}[X_n \mathbf{1}_{\{|X_n|>1\}}]| \le \mathbb{E}[|X_n| \mathbf{1}_{\{|X_n|>1\}}],$$

and the right side is summable. All three conditions hold, so $\sum_n X_n$ converges a.s.

Exercise 2.5.7

Question. Let X_n be independent. Suppose

$$\sum_n \mathbb{E}|X_n|^{p(n)} < \infty, \qquad 0 < p(n) \le 2,$$

and $\mathbb{E}X_n = 0$ when $p(n) > 1$. Show that $\sum_n X_n$ converges a.s.

Knowledge used. `durrett_ch2_three_series_theorem`; `durrett_ch2_truncation_method`.

Solution. For all n,

$$\mathbb{P}(|X_n| > 1) \le \mathbb{E}|X_n|^{p(n)},$$

so large jumps occur only finitely often a.s. If $p(n) \le 1$, then on $\{|X_n| \le 1\}$ we have $|X_n| \le |X_n|^{p(n)}$, so the corresponding small-jump parts are absolutely summable in expectation, hence a.s.

For the indices with $p(n) > 1$, apply the three-series theorem. On $|x| \le 1$, $x^2 \le |x|^{p(n)}$ because $p(n) \le 2$, and on $|x| > 1$, $|x| \le |x|^{p(n)}$. Hence

$$\sum_{p(n)>1} \operatorname{var}(X_n \mathbf{1}_{\{|X_n| \le 1\}}) < \infty$$

and, using $\mathbb{E}X_n = 0$,

$$\sum_{p(n)>1} |\mathbb{E}[X_n \mathbf{1}_{\{|X_n| \le 1\}}]| \le \sum_{p(n)>1} \mathbb{E}[|X_n| \mathbf{1}_{\{|X_n|>1\}}] < \infty.$$

Thus the full series converges a.s.

Exercise 2.5.8

Question. Let X_n be i.i.d. and not identically 0. For the power series $\sum_{n\geq 1} X_n(\omega)z^n$, show that its radius of convergence is 1 a.s. if $\mathbb{E}\log^+|X_1| < \infty$, and is 0 a.s. if $\mathbb{E}\log^+|X_1| = \infty$.

Knowledge used. `durrett_ch2_borel_cantelli`; `durrett_ch2_tail_sum_moment_test`; `durrett_ch2_power_series_radius`.

Solution. The radius is

$$r(\omega) = \frac{1}{\limsup_n |X_n(\omega)|^{1/n}}.$$

If $\mathbb{E}\log^+|X_1| < \infty$, then for every $\varepsilon > 0$,

$$\sum_n \mathbb{P}(\log^+|X_n| > \varepsilon n) < \infty.$$

Thus eventually $|X_n|^{1/n} \leq e^{\varepsilon}$, and the limsup is at most 1. Since X_1 is not identically 0, there is $\delta > 0$ with $\mathbb{P}(|X_1| > \delta) > 0$; by Borel–Cantelli this occurs infinitely often, so the limsup is at least 1. Hence $r = 1$.

If $\mathbb{E}\log^+|X_1| = \infty$, then for every $M > 1$,

$$\sum_n \mathbb{P}(\log^+|X_n| > n\log M) = \infty.$$

By Borel–Cantelli, $|X_n| > M^n$ infinitely often, so $\limsup_n |X_n|^{1/n} \geq M$. Since M is arbitrary, the limsup is infinite and $r = 0$.

Exercise 2.5.9

Question. Let X_n be independent and $S_{m,n} = X_{m+1} + \cdots + X_n$. Prove

$$\mathbb{P}\left(\max_{m<j\leq n} |S_{m,j}| > 2a\right) \min_{m<k\leq n} \mathbb{P}(|S_{k,n}| \leq a) \leq \mathbb{P}(|S_{m,n}| > a).$$

Knowledge used. `durrett_ch2_levy_maximal_inequality`; `durrett_ch2_independent_increments`.

Solution. Let A_j be the event that j is the first index in $(m,n]$ for which $|S_{m,j}| > 2a$. The events A_j are disjoint and A_j depends only on $X_{m+1},\ldots,X_j$, while $S_{j,n}$ depends only on later variables. If A_j and $|S_{j,n}| \leq a$ occur, then

$$|S_{m,n}| = |S_{m,j} + S_{j,n}| > a.$$

Therefore

$$\mathbb{P}(A_j)\mathbb{P}(|S_{j,n}| \leq a) = \mathbb{P}(A_j \cap \{|S_{j,n}| \leq a\}) \leq \mathbb{P}(A_j \cap \{|S_{m,n}| > a\}).$$

Summing over j and taking the minimum over k proves the claim.

Exercise 2.5.10

Question. Use the preceding inequality to prove Levy's theorem: if $S_n = \sum_{i=1}^n X_i$ with independent X_i and $\lim_n S_n$ exists in probability, then $\lim_n S_n$ exists a.s.

Knowledge used. `durrett_ch2_levy_maximal_inequality`; `durrett_ch2_cauchy_in_probability`; `durrett_ch2_borel_cantelli`.

Solution. Since S_n converges in probability, it is Cauchy in probability: for every $a > 0$,

$$\sup_{k,n \geq m} \mathbb{P}(|S_n - S_k| > a) \to 0.$$

Applying Exercise 2.5.9 to the block $(m, n]$, for large m the minimum factor is at least $1/2$, hence

$$\mathbb{P}\left(\max_{m<j\leq n} |S_j - S_m| > 2a\right) \leq 2\mathbb{P}(|S_n - S_m| > a).$$

Thus the partial sums are Cauchy in probability uniformly over each tail maximum. Choose $m_r \uparrow \infty$ so that

$$\mathbb{P}\left(\max_{m_r<j\leq m_{r+1}} |S_j - S_{m_r}| > 2^{-r}\right) < 2^{-r}.$$

Borel–Cantelli implies that, a.s., only finitely many of these block oscillations exceed 2^{-r}. Since S_{m_r} is Cauchy a.s. along the chosen subsequence, the full sequence S_n is Cauchy a.s. and therefore converges a.s.

Exercise 2.5.11

Question. Let X_i be i.i.d. and $S_n = \sum_{i=1}^n X_i$. Use the inequality in Exercise 2.5.9 to show that if $S_n/n \to 0$ in probability, then

$$\frac{1}{n} \max_{1\leq m\leq n} S_m \to 0 \qquad \text{in probability.}$$

Knowledge used. `durrett_ch2_levy_maximal_inequality`; `durrett_ch2_convergence_in_probability`.

Solution. It is enough to prove the stronger statement with $\max |S_m|$. Put $m = 0$ in Exercise 2.5.9. For fixed $\varepsilon > 0$,

$$\mathbb{P}\left(\max_{1\leq j\leq n} |S_j| > 2\varepsilon n\right) \min_{1\leq k\leq n} \mathbb{P}(|S_n - S_k| \leq \varepsilon n) \leq \mathbb{P}(|S_n| > \varepsilon n).$$

Since $S_n/n \to 0$ in probability, the right side tends to 0. Also $S_n - S_k$ has the same law as S_{n-k}, and

$$\min_{0\leq \ell\leq n} \mathbb{P}(|S_\ell| \leq \varepsilon n) \to 1;$$

for large ℓ this follows from $|S_\ell|/\ell \to 0$ in probability, and for finitely many small ℓ it is immediate because $\varepsilon n \to \infty$. Hence

$$n^{-1} \max_{1\leq j\leq n} |S_j| \to 0 \quad \text{in probability,}$$

which implies the desired one-sided result.

Exercise 2.5.12

Question. Let X_i be i.i.d., $S_n = \sum_{i=1}^n X_i$, and let $a(n) \uparrow \infty$ with $a(2^n)/a(2^{n-1})$ bounded.

(i) Use Exercise 2.5.9 to show that if $S_n/a(n) \to 0$ in probability and $S_{2^n}/a(2^n) \to 0$ a.s., then $S_n/a(n) \to 0$ a.s.

(ii) If $\mathbb{E}X_1 = 0$ and $\mathbb{E}X_1^2 < \infty$, conclude that for every $\varepsilon > 0$,

$$\frac{S_n}{n^{1/2}(\log_2 n)^{1/2+\varepsilon}} \to 0 \qquad \text{a.s.}$$

Knowledge used. durrett_ch2_levy_maximal_inequality; durrett_ch2_chebyshev; durrett_ch2_borel_cantelli.

Solution. (i) Write $b_k = a(2^k)$. On the dyadic block $2^{k-1} < j \le 2^k$, Exercise 2.5.9 gives, for fixed $\eta > 0$,

$$\mathbb{P}\left(\max_{2^{k-1}<j\le 2^k} |S_j - S_{2^{k-1}}| > 2\eta b_k\right) \le \frac{\mathbb{P}(|S_{2^k} - S_{2^{k-1}}| > \eta b_k)}{q_k},$$

where

$$q_k = \min_{2^{k-1}<j\le 2^k} \mathbb{P}(|S_{2^k} - S_j| \le \eta b_k).$$

The assumption $S_n/a(n) \to 0$ in probability and the boundedness of $a(2^k)/a(2^{k-1})$ imply $q_k \to 1$. The dyadic convergence $S_{2^k}/b_k \to 0$ a.s. makes the endpoint increments negligible; the standard Borel–Cantelli blocking argument from Levy's theorem then gives

$$\max_{2^{k-1}<j\le 2^k} \frac{|S_j - S_{2^{k-1}}|}{b_k} \to 0 \qquad \text{a.s.}$$

Since $a(j)$ is increasing and $b_k/a(j)$ is bounded on the block, and since $S_{2^{k-1}}/a(2^{k-1}) \to 0$ a.s., it follows that $S_n/a(n) \to 0$ a.s.

(ii) Let

$$a(n) = n^{1/2}(\log_2 n)^{1/2+\varepsilon}, \qquad n \ge 2.$$

Then $a(2^k) = 2^{k/2}k^{1/2+\varepsilon}$ and

$$\sum_{k=1}^{\infty} \mathbb{P}(|S_{2^k}| > \delta a(2^k)) \le \sum_{k=1}^{\infty} \frac{\operatorname{var}(S_{2^k})}{\delta^2 a(2^k)^2} = \frac{\operatorname{var}(X_1)}{\delta^2} \sum_{k=1}^{\infty} \frac{1}{k^{1+2\varepsilon}} < \infty.$$

Borel–Cantelli yields $S_{2^k}/a(2^k) \to 0$ a.s. Chebyshev also gives $S_n/a(n) \to 0$ in probability. Part (i) completes the proof.

Exercise 2.6.1

Question. For a renewal process with interarrival times ξ_i, renewal count N_t, and renewal function $U(t) = \mathbb{E}N_t$, prove

$$\frac{t}{\mathbb{E}(\xi_1 \wedge t)} \le U(t) \le \frac{2t}{\mathbb{E}(\xi_1 \wedge t)}.$$

Knowledge used. durrett_ch2_renewal_function; durrett_ch2_wald_identity_truncated.

Solution. Let $\eta_i = \xi_i \wedge t$. Since $T_{N_t} > t$, while each $\eta_i \le t$,

$$t \le \sum_{i=1}^{N_t} \eta_i \le 2t.$$

Indeed, the partial sum before the last term is at most t, and the last truncated term is at most t. Wald's identity for the bounded increments η_i gives

$$\mathbb{E}\sum_{i=1}^{N_t} \eta_i = \mathbb{E}(\xi_1 \wedge t)\,\mathbb{E}N_t.$$

Dividing by $\mathbb{E}(\xi_1 \wedge t)$ proves the two inequalities.

Exercise 2.6.2

Question. Deduce the elementary renewal theorem from the strong law by showing

$$\limsup_{t\to\infty} \mathbb{E}(N_t/t)^2 < \infty.$$

Knowledge used. `durrett_ch2_elementary_renewal_theorem`; `durrett_ch2_renewal_slln`; `durrett_ch2_uniform_integrability`.

Solution. Choose $\delta > 0$ with $p = \mathbb{P}(\xi_1 > \delta) > 0$, and let $K = \lceil t/\delta \rceil$. If $N_t > mK$, then among the first mK interarrival times no block of length K can consist entirely of variables larger than δ. Thus

$$\mathbb{P}(N_t > mK) \le (1 - p^K)^m.$$

This geometric tail bound implies $\mathbb{E}N_t^2 \le CK^2 \le C't^2$, so $\limsup_t \mathbb{E}(N_t/t)^2 < \infty$.

By the strong law for renewal times, $N_t/t \to 1/\mu$ a.s. when $\mu = \mathbb{E}\xi_1 < \infty$. The L^2 bound gives uniform integrability, so $\mathbb{E}(N_t/t) \to 1/\mu$, i.e. $U(t)/t \to 1/\mu$.

Exercise 2.6.3

Question. Poisson arrivals occur at rate 1. A server accepts a customer iff the server is free, and service times have distribution F with mean μ. Show that accepted arrival times form a renewal process with mean interarrival time $\mu + 1$, and that the limiting fraction of arrivals served is $1/(\mu + 1)$.

Knowledge used. `durrett_ch2_poisson_memoryless`; `durrett_ch2_renewal_reward_theorem`; `durrett_ch2_renewal_slln`.

Solution. After an accepted customer begins service, the server is unavailable for a service time with mean μ. Once service ends, the memoryless property of the Poisson process says the waiting time to the next arrival is exponential with mean 1, independent of the past. Hence accepted arrivals form a renewal process with interarrival law

$$\text{service time} + \text{Exp}(1),$$

whose mean is $\mu + 1$. Therefore the accepted arrival rate is $1/(\mu + 1)$. The total arrival rate is 1, so the limiting fraction served is $1/(\mu + 1)$.

Exercise 2.6.4

Question. Let $A_t = t - T_{N(t)-1}$ be the age. For fixed $x > 0$, show that $H(t) = \mathbb{P}(A_t > x)$ satisfies

$$H(t) = (1 - F(t))\mathbf{1}_{(x,\infty)}(t) + \int_0^t H(t-s)\,dF(s),$$

and conclude

$$\mathbb{P}(A_t > x) \to \frac{1}{\mu}\int_x^\infty (1 - F(u))\,du.$$

Knowledge used. durrett_ch2_renewal_equation; durrett_ch2_key_renewal_theorem; durrett_ch2_age_residual_life.

Solution. Condition on the first renewal time T_1. If $T_1 > t$, then $A_t = t$, so the event occurs exactly when $t > x$. If $T_1 = s \le t$, the process restarts at time s, and the new age at time t has the same law as A_{t-s}. This gives the stated renewal equation.

The forcing function is

$$h(t) = (1 - F(t))\mathbf{1}_{(x,\infty)}(t),$$

which is directly Riemann integrable when $\mu < \infty$. The key renewal theorem yields

$$H(t) \to \frac{1}{\mu}\int_0^\infty h(u)\,du = \frac{1}{\mu}\int_x^\infty (1 - F(u))\,du.$$

Exercise 2.6.5

Question. Use the renewal equation in Exercise 2.6.4 and Theorem 2.6.9 to show that if the renewal process is a rate λ Poisson process, then A_t has the same distribution as $\xi_1 \wedge t$.

Knowledge used. durrett_ch2_poisson_memoryless; durrett_ch2_renewal_equation;

Solution. For Poisson renewals, $F(u) = 1 - e^{-\lambda u}$. For $0 < x < t$,

$$\mathbb{P}(A_t > x) = \mathbb{P}(\text{no renewal in } (t - x, t]) = e^{-\lambda x},$$

and for $x \ge t$ this probability is 0. This is also

$$\mathbb{P}(\xi_1 \wedge t > x),$$

since ξ_1 is exponential with rate λ. Thus $A_t \overset{d}{=} \xi_1 \wedge t$.

Exercise 2.6.6

Question. Let $B_t = T_{N(t)} - t$ be the residual lifetime. Show that

$$\mathbb{P}(A_t > x,\ B_t > y) \to \frac{1}{\mu}\int_{x+y}^\infty (1 - F(u))\,du.$$

Knowledge used. durrett_ch2_age_residual_life; durrett_ch2_key_renewal_theorem;

Solution. The event $\{A_t > x, B_t > y\}$ means that the renewal interval straddling t began before $t - x$ and has length exceeding $x + y$. Its renewal equation has forcing term

$$h(t) = (1 - F(t + y))\mathbf{1}_{(x,\infty)}(t).$$

By the key renewal theorem,

$$\mathbb{P}(A_t > x, B_t > y) \to \frac{1}{\mu}\int_0^\infty h(u)\,du = \frac{1}{\mu}\int_x^\infty (1 - F(u + y))\,du = \frac{1}{\mu}\int_{x+y}^\infty (1 - F(v))\,dv.$$

Exercise 2.6.7

Question. In an alternating renewal process, working times have distribution F_1 with mean μ_1, repair times have distribution F_2 with mean μ_2, and one cycle has law $F = F_1 * F_2$. If F is nonarithmetic, show that the probability $H(t)$ that the machine is working at time t satisfies

$$H(t) \to \frac{\mu_1}{\mu_1 + \mu_2}.$$

Knowledge used. `durrett_ch2_alternating_renewal_process`; `durrett_ch2_key_renewal_theorem`; `durrett_ch2_renewal_reward_theorem`.

Solution. The machine is working at time t either because the initial working period exceeds t, or because a full cycle has ended and the restarted process is working. Thus

$$H(t) = 1 - F_1(t) + \int_0^t H(t-s)\, dF(s).$$

The forcing function $h(t) = 1 - F_1(t)$ is directly Riemann integrable and

$$\int_0^\infty h(t)\, dt = \mu_1.$$

The mean cycle length is $\mu_1 + \mu_2$. The key renewal theorem gives

$$H(t) \to \frac{\mu_1}{\mu_1 + \mu_2}.$$

Exercise 2.6.8

Question. Find a renewal equation for $H(t) = \mathbb{P}(\text{the number of renewals in } [0,t] \text{ is odd})$, and use the renewal theorem to show that $H(t) \to 1/2$.

Knowledge used. `durrett_ch2_renewal_equation`; `durrett_ch2_alternating_renewal_process`; `durrett_ch2_key_renewal_theorem`.

Solution. Counting the renewal at time 0, if the first interarrival exceeds t, the count is 1 and hence odd. If the first interarrival is $s \le t$, parity is reversed relative to the restarted process. Hence

$$H(t) = 1 - F(t) + \int_0^t (1 - H(t-s))\, dF(s).$$

This is the alternating-renewal equation with equal on- and off-cycle distributions. By Exercise 2.6.7, the limiting fraction of time spent in either parity is

$$\frac{\mu}{\mu + \mu} = \frac{1}{2}.$$

Thus $H(t) \to 1/2$.

Exercise 2.6.9

Question. Assume F has directly Riemann integrable density f. Let $V = U - \mathbf{1}_{[0,\infty)}$. Show that V has density v satisfying

$$v(t) = f(t) + \int_0^t v(t-s)\, dF(s),$$

and conclude that $v(t) \to 1/\mu$.
Knowledge used. durrett_ch2_renewal_density; durrett_ch2_key_renewal_theorem.
Solution. Since

$$U = \sum_{n=0}^{\infty} F^{*n}, \qquad V = \sum_{n=1}^{\infty} F^{*n},$$

the measure V has density

$$v(t) = \sum_{n=1}^{\infty} f^{*n}(t).$$

Separating the first term gives

$$v(t) = f(t) + \sum_{n=2}^{\infty} f^{*n}(t) = f(t) + \int_0^t v(t-s)\,dF(s).$$

The key renewal theorem applied to the directly Riemann integrable function f gives

$$v(t) \to \frac{1}{\mu}\int_0^{\infty} f(s)\,ds = \frac{1}{\mu}.$$

Exercise 2.7.1

Question. For $\gamma(a)$ in the large-deviation theorem, show that the following are equivalent:

$$\gamma(a) = -\infty, \qquad \mathbb{P}(X_1 \ge a) = 0, \qquad \mathbb{P}(S_n \ge na) = 0 \text{ for all } n.$$

Knowledge used. durrett_ch2_large_deviation_gamma; durrett_ch2_independent_sums.
Solution. If $\mathbb{P}(X_1 \ge a) = 0$, then every summand is $< a$ a.s.; hence $S_n < na$ a.s. and $\mathbb{P}(S_n \ge na) = 0$ for all n, so $\gamma(a) = -\infty$.

Conversely, if $\mathbb{P}(S_n \ge na) = 0$ for all n, taking $n = 1$ gives $\mathbb{P}(X_1 \ge a) = 0$. Finally, if $\gamma(a) = -\infty$ but $p = \mathbb{P}(X_1 \ge a) > 0$, then $\mathbb{P}(S_n \ge na) \ge p^n$, so $n^{-1}\log\mathbb{P}(S_n \ge na) \ge \log p > -\infty$, a contradiction.

Exercise 2.7.2

Question. Use the definition of γ to show that for rational $\lambda \in [0,1]$,

$$\gamma(\lambda a + (1-\lambda)b) \ge \lambda\gamma(a) + (1-\lambda)\gamma(b).$$

Extend this to all λ and show that γ is concave, hence Lipschitz continuous on compact subsets of $\{\gamma > -\infty\}$.
Knowledge used. durrett_ch2_large_deviation_gamma; durrett_ch2_superadditivity; durrett_ch2_concavity.
Solution. Let $\lambda = r/(r+s)$ with integers $r, s \ge 0$. Independence gives

$$\mathbb{P}(S_{r+s} \ge ra + sb) \ge \mathbb{P}(S_r \ge ra)\mathbb{P}(S'_s \ge sb),$$

where S'_s is an independent copy of S_s. Iterating this block argument and taking logarithms divided by the total length yields

$$\gamma\left(\frac{ra+sb}{r+s}\right) \ge \frac{r}{r+s}\gamma(a) + \frac{s}{r+s}\gamma(b).$$

Thus the inequality holds for rational λ. Since γ is nonincreasing in a, approximating an arbitrary λ from above and below by rationals extends the inequality to all $\lambda \in [0,1]$. Therefore γ is concave on its effective domain. A finite concave function on an open interval has bounded one-sided slopes on compact subintervals, hence is Lipschitz there.

Exercise 2.7.3

Question. Let X_i be i.i.d. Poisson random variables with mean 1. Find

$$\lim_{n\to\infty} \frac{1}{n} \log \mathbb{P}(S_n \geq na), \qquad a > 1.$$

Knowledge used. durrett_ch2_cramer_theorem; durrett_ch2_poisson_large_deviation; durrett_ch2_chernoff_bound.

Solution. Here S_n is Poisson with mean n. The moment generating function of X_1 is

$$\phi(\theta) = \exp(e^\theta - 1).$$

For $a > 1$, the optimizing Chernoff parameter solves $\phi'(\theta)/\phi(\theta) = e^\theta = a$, so $\theta = \log a$. The rate function is

$$I(a) = a \log a - a + 1.$$

Therefore

$$\lim_{n\to\infty} \frac{1}{n} \log \mathbb{P}(S_n \geq na) = -I(a) = -a \log a + a - 1.$$

Exercise 2.7.4

Question. For fair coin flips written as $X_i = \pm 1$, prove

$$\phi(\theta) \leq \exp(\phi(\theta) - 1) \leq \exp(\beta\theta^2), \qquad \theta \leq 1,$$

where

$$\beta = \sum_{n=1}^{\infty} \frac{1}{(2n)!} \approx 0.586,$$

and use Chernoff's inequality to show that for $0 \leq a \leq 1$,

$$\mathbb{P}(S_n \geq an) \leq \exp\left(-\frac{na^2}{4\beta}\right).$$

Knowledge used. durrett_ch2_chernoff_bound; durrett_ch2_mgf_bounds; durrett_ch2_subgaussian_coin_flips.

Solution. For $X_1 = \pm 1$ with probability $1/2$,

$$\phi(\theta) = \mathbb{E}e^{\theta X_1} = \cosh\theta.$$

Since $x \leq e^{x-1}$ for $x > 0$,

$$\phi(\theta) \leq e^{\phi(\theta)-1}.$$

For $0 \leq \theta \leq 1$,

$$\phi(\theta) - 1 = \sum_{n=1}^{\infty} \frac{\theta^{2n}}{(2n)!} \leq \theta^2 \sum_{n=1}^{\infty} \frac{1}{(2n)!} = \beta\theta^2.$$

Thus $\phi(\theta) \le e^{\beta\theta^2}$. Chernoff's bound gives

$$\mathbb{P}(S_n \ge an) \le \exp\{-n(a\theta - \beta\theta^2)\}.$$

For $0 \le a \le 1$, choose $\theta = a/(2\beta) \le 1$. Then $a\theta - \beta\theta^2 = a^2/(4\beta)$, so

$$\mathbb{P}(S_n \ge an) \le \exp\left(-\frac{na^2}{4\beta}\right).$$

Exercise 2.7.5

Question. Suppose $\mathbb{E}X_i = 0$ and $\mathbb{E}e^{\theta X_i} = \infty$ for all $\theta > 0$. Show that for every $a > 0$,

$$\frac{1}{n}\log \mathbb{P}(S_n \ge na) \to 0.$$

Knowledge used. durrett_ch2_heavy_tail_large_deviation; durrett_ch2_one_big_jump; durrett_ch2_large_deviation_gamma.

Solution. The limit $\gamma(a)$ exists by the large-deviation theorem and is at most 0. Suppose $\gamma(a) < 0$ for some $a > 0$. Then for some $c > 0$,

$$\mathbb{P}(S_n \ge na) \le e^{-cn}$$

for all large n. By the weak law, for large n,

$$\mathbb{P}(S_{n-1} \ge -na) \ge \frac{1}{2}.$$

Therefore

$$\mathbb{P}(X_1 \ge 2na) \le 2\mathbb{P}(S_n \ge na) \le 2e^{-cn}.$$

Monotonicity fills the gaps between multiples of $2a$, so the positive tail of X_1 is exponentially bounded. This would imply $\mathbb{E}e^{\theta X_1} < \infty$ for some $\theta > 0$, contradicting the assumption. Hence $\gamma(a) = 0$ for every $a > 0$.

Exercise 2.7.6

Question. Show that if $\mathbb{E}X_i = 0$, then for $a, \varepsilon > 0$,

$$\liminf_{n\to\infty} \frac{\mathbb{P}(S_n \ge na)}{n\mathbb{P}(X_1 \ge n(a+\varepsilon))} \ge 1.$$

Knowledge used. durrett_ch2_one_big_jump; durrett_ch2_weak_law; durrett_ch2_tail_probability.

Solution. Let $p_n = \mathbb{P}(X_1 \ge n(a+\varepsilon))$. Since $\mathbb{E}|X_1| < \infty$, $np_n \to 0$. Consider the event that exactly one of $X_1, \dots, X_n$ is at least $n(a+\varepsilon)$ and the remaining sum is at least $-n\varepsilon$. On this event, $S_n \ge na$. Independence and the weak law give

$$\mathbb{P}(S_{n-1} \ge -n\varepsilon) \to 1.$$

Hence

$$\mathbb{P}(S_n \ge na) \ge np_n(1-p_n)^{n-1}\mathbb{P}(S_{n-1} \ge -n\varepsilon),$$

and the right side is asymptotic to np_n. Dividing by $n\mathbb{P}(X_1 \ge n(a+\varepsilon))$ proves the desired lower bound.